Blue Impulse — Markdown Content V2.0

Module: Hardware, Networking and Behaviorist AI — How Machines Sense, Connect and Act
Team: Blue Impulse
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Start here

A small campus delivery rover, Pip, must sense the world, choose useful actions and execute them on real hardware. The rover is fictional; its message identifier in the worked protocol examples is robot-7. Its problems connect the six chapters without claiming that every smart device learns or that a successful demo proves a safe deployment.

You need basic arithmetic and a willingness to record predictions. No programming experience, online account or paid tool is required. Paper versions contain every input and expected result. Optional code requires Python 3.9 or newer and uses only the standard library.

Read each chapter's outcomes, then one topic at a time. Study the worked explanation, attempt its application checkpoint, compare the feedback and explain any difference. Complete the chapter mission before the exercises, then use the separate solutions to review your reasoning. Use the glossary when a term is unfamiliar. The separate Test Bank places all questions before the answer key.

Learning route and outcomes

Chapter	Learning outcome	Mission
1 — The Fundamentals	Represent numbers, characters, images and sound; explain stored programs, registers and the fetch–execute cycle; compare translators; describe operating-system tasks, utilities, interrupts and peripheral connections.	ACT-01 — The parcel total is wrong
2 — Behaviorism & Behaviorist AI	Distinguish classical and operant conditioning; identify stimulus–response rules and layered robotics; explain reinforcement-learning components, policies, values and exploration; interpret a Q update and the parts of an MDP.	ACT-02 — One lucky route is not a policy
3 — Hardware	Classify hardware; compare memory and storage; explain processors, pipelining, parallelism and interrupts; connect sensors, converters and actuators; reason with logic gates; evaluate embedded systems, interfaces and AI accelerators.	ACT-03 — The kitchen with four stations
4 — Networking	Compare network architectures, clients and clouds; identify communication hardware and media; trace URLs, DNS and TCP/IP; explain application protocols, switching and network performance; distinguish data protection goals, encryption, signatures and TLS.	ACT-04 — The fast network sends a late warning
5 — Applications	Trace an integrated smart-system workflow; explain real-time deadlines and edge/cloud choices; connect policies to actuators; evaluate automation, human-AI interaction, ethical concerns and connected applications.	ACT-05 — Yesterday's command arrives today
6 — Strengths & Limitations	Evaluate physical and network constraints; distinguish scalability, interoperability and resilience; compare predictive analytics with behavioural modelling; assess resource allocation, explainability, bias and responsible deployment.	ACT-06 — Choose a system, not a leaderboard

What is included

Every chapter contains developed explanations, diagrams expressed as readable flows/tables, an application checkpoint with feedback for each topic, a substantial mission, a glossary, exercises, answers and specific references. Source notes distinguish textbook-derived material, technical standards, fictional examples, team-provided art and AI-assisted revision.

The original guide source supplied the existing developed explanations. On 7 November they were reconciled into this shared Markdown foundation; future edits should start here and update the guide, slides and tutorials. This is a documented repair of a missing Markdown foundation, not a claim that the earlier production already followed the desired sequence.

Navigation

- Chapter 1: The Fundamentals
- Chapter 2: Behaviorism & Behaviorist AI
- Chapter 3: Hardware
- Chapter 4: Networking
- Chapter 5: Applications
- Chapter 6: Strengths & Limitations
- Curated resources and attribution
- Activity setup and expected results
- Review status and evidence

Review status

Automated checks and AI-assisted content review are recorded in the supplied evidence. A teammate must still independently attempt every Test Bank question and selected activities. No teacher feedback, learner test or student reflection has been invented. The remaining human review requirement is explicit in the submission report.

Chapter 1 — The Fundamentals

Learning outcome: Represent numbers, characters, images and sound; explain stored programs, registers and the fetch–execute cycle; compare translators; describe operating-system tasks, utilities, interrupts and peripheral connections.

Prerequisites: Basic arithmetic; Chapter 1 supplies representation and execution foundations.

From bits to bytes

Two states, many meanings

A bit is a binary digit: 0 or 1. Electronic circuits distinguish two ranges of voltage, allowing hardware to store and process bits reliably. Eight bits form one byte. A word is the group of bits a processor handles as a unit; its size depends on the architecture. The same pattern may represent a number, a character or an instruction: the interpretation supplies the meaning.

Measure storage accurately

Prefix	Bytes
kB / MB / GB	1,000 / 1,000,000 / 1,000,000,000
KiB / MiB	1,024 / 1,048,576
GiB / TiB	1,073,741,824 / 1,099,511,627,776

Units matter

Lowercase b means bit; uppercase B means byte. Divide bits by 8 to obtain bytes. Decimal prefixes use powers of ten; binary prefixes use powers of two. For example, 2 KiB = 2 × 1,024 = 2,048 bytes. A network speed in Mb/s is not the same unit as a file size in MB.

Apply and check

Try before reading feedback: A log is 2 KiB. How many bits must be transferred, before protocol overhead?

Feedback: 2 × 1,024 × 8 = 16,384 bits. Keep binary storage prefixes separate from decimal link-rate prefixes.

Practice connection: Exercise 1.2b3 in exercises.md; explained answer in solutions.md.

Binary, denary & hex

Place value does the work

Denary uses powers of 10. Binary uses powers of 2, so the positions in an eight-bit number are 128, 64, 32, 16, 8, 4, 2 and 1. Add the positions containing 1: 00101101 = 32 + 8 + 4 + 1 = 45. To convert denary to binary, select those powers of two, or repeatedly divide by 2 and read the remainders backwards.

Hexadecimal groups four bits

Binary nibble	Hex	Denary
0000–1001	0–9	0–9
1010 / 1011 / 1100	A / B / C	10 / 11 / 12
1101 / 1110 / 1111	D / E / F	13 / 14 / 15

Why engineers use hexadecimal

Group binary digits in fours from the right: 0010 1101 becomes 2D. Hexadecimal is a compact notation for the same value, useful in memory dumps, colour codes and hardware addresses. Binary addition carries when a column reaches 2: 1 + 1 = 10. With a fixed width, check whether a carry or signed overflow changes the interpretation.

Apply and check

Try before reading feedback: Pip reports status 0x2D. Express the same unsigned value in binary and denary.

Feedback: 0010 1101 and 45; hexadecimal changes notation, not the stored value.

Practice connection: Exercise 1.1a1 in exercises.md; explained answer in solutions.md.

Represent negative values

Three eight-bit representations of −5

Method	Representation	Rule
Sign and magnitude	10000101	Sign bit + magnitude
One’s complement	11111010	Invert the bits of +5
Two’s complement	11111011	Invert +5, then add 1

Read two’s complement

The leftmost bit has a negative weight. In eight bits, 11111011 means −128 + 64 + 32 + 16 + 8 + 2 + 1 = −5. In n bits, the range is −2^(n−1) to 2^(n−1) − 1. Always state the width: a bit pattern has no unique signed meaning without a representation and a width.

Subtract by adding the negative

To calculate 9 − 3 in eight bits, add 00001001 to the two’s complement of 00000011: 11111101. Keep the eight result bits, giving 00000110. A carry outside the width is discarded. Signed overflow occurs when the true result lies outside the representable range; it is not the same as an unsigned carry.

Apply and check

Try before reading feedback: An eight-bit signed sensor adds 100 and 40. Can the exact sum fit?

Feedback: No. The true sum 140 exceeds +127, so eight-bit signed overflow occurs. Do not reinterpret the wrapped result as the true physical value.

Practice connection: Exercise 1.1a3 in exercises.md; explained answer in solutions.md.

Numbers, characters & codes

BCD stores one decimal digit at a time

Binary-coded decimal allocates four bits to each decimal digit. Decimal 259 becomes 0010 0101 1001. This is different from converting the whole number 259 to ordinary binary. Patterns 1010–1111 are unused as decimal digits in ordinary BCD. Decimal representations are useful where exact decimal digits matter, including displays and some financial calculations.

Characters need an agreed code

A character set defines the available characters. An encoding maps characters to stored values. ASCII defines 128 codes, covering English letters, digits, punctuation and control characters. Unicode assigns code points to characters from many writing systems. UTF-8 and UTF-16 are ways to encode those code points; Unicode does not mean every character occupies the same number of bytes.

One pattern, different interpretations

Step / element	Meaning in the diagram
1	Bits
2	Encoding / format
3	Meaning

An application must know the agreed format before it can interpret stored data.

Apply and check

Try before reading feedback: A display sends digits 2 and 5 as BCD. Is 0010 0101 the ordinary binary value 25?

Feedback: No. The nibbles encode decimal digits 2 and 5. As one unsigned binary byte, the same pattern is 37.

Practice connection: Exercise 1.1b2 in exercises.md; explained answer in solutions.md.

Build a digital image

Pixels and colour depth

A bitmap is a grid of pixels. Pixel dimensions describe its width and height; resolution may also describe pixel density when an image is displayed or printed. Colour depth is the number of bits used per pixel. With b bits per pixel, up to 2^b different codes are available. More pixels preserve finer spatial detail; more bits per pixel allow more colour or intensity values.

Uncompressed bitmap size

Quantity	Example
Dimensions	1,000 × 500 pixels
Colour depth	8 bits per pixel
Raw pixel data	1,000 × 500 × 8 = 4,000,000 bits
In bytes	4,000,000 ÷ 8 = 500,000 bytes

Bitmap versus vector

A vector image stores shapes, paths and their properties. It can scale without pixelation, making it useful for diagrams and logos. Photographs are normally bitmaps. The size formula above excludes headers, palettes, metadata and compression; an actual file may have a different size.

Apply and check

Try before reading feedback: Halve both dimensions of a 1,000 × 500 eight-bit bitmap. What happens to raw pixel size?

Feedback: 500 × 250 × 1 byte = 125,000 bytes, one quarter of 500,000. Halving two dimensions quarters the pixel count.

Practice connection: Exercise 1.2b2 in exercises.md; explained answer in solutions.md.

Sample, store & compress

Turn an analogue wave into numbers

An analogue-to-digital converter samples a sound signal. Sampling rate is the number of samples per second, in hertz. Sampling resolution, or bit depth, is the number of bits per sample. A higher rate can represent a wider range of frequencies; a greater bit depth provides finer amplitude steps. Both increase uncompressed storage requirements.

Estimate audio size

Uncompressed audio size in bits = sampling rate × bit depth × channels × duration. For 10 seconds of mono audio at 8,000 samples/s and 16 bits/sample: 8,000 × 16 × 1 × 10 = 1,280,000 bits, or 160,000 bytes. Stereo requires two channels.

Choose the right compression

Lossless	Lossy
Reconstructs the original data exactly.	Permanently discards some information.
ZIP, run-length encoding, PNG.	JPEG images, MP3 audio.
Suitable for programs and exact records.	Useful when smaller media files justify reduced fidelity.

Apply and check

Try before reading feedback: Keep the duration and bit depth fixed but change mono to stereo. Predict raw audio size.

Feedback: It doubles because channels change from 1 to 2. Compression and headers can change the actual file size.

Practice connection: Exercise 1.2a2 in exercises.md; explained answer in solutions.md.

The stored-program computer

Instructions and data share memory

In the introductory Von Neumann model, memory holds both instructions and data. The CPU fetches instructions, decodes them and performs the required operations. The arithmetic logic unit (ALU) performs calculations and comparisons. The control unit coordinates the operation of the CPU, memory and input/output devices. Registers hold small amounts of immediately needed information.

The main connections

Step / element	Meaning in the diagram
1	CPU
2	System buses
3	Memory / I/O

Address bus: identifies a location. Data bus: carries instructions or values. Control bus: carries signals such as read and write.

Timing and the bottleneck

The system clock coordinates operations. A clock frequency of 3 GHz means three billion clock cycles per second, not necessarily three billion completed instructions. Cache and pipelining reduce some delays, but moving instructions and data through shared resources can still restrict throughput. Real processors extend this simplified model in many ways.

Apply and check

Try before reading feedback: Pip has a 3 GHz processor. A friend says it therefore finishes exactly three billion instructions each second. Find the error.

Feedback: GHz counts clock cycles, not completed instructions. Work per instruction, parallel issue, stalls and workload all affect instruction throughput.

Practice connection: Exercise 1.3 in exercises.md; explained answer in solutions.md.

Fetch / decode / execute

Follow one instruction

Step	Register transfer or action
1	Copy the address in PC to MAR.
2	Read the memory instruction into MDR.
3	Advance PC to the next instruction address.
4	Copy the instruction from MDR to CIR.
5	Decode the operation and its operand.
6	Execute; store a result or change control flow.

Give each register a job

PC is the program counter: the next instruction address. MAR identifies the memory location being accessed. MDR holds data moving to or from memory. CIR holds the current instruction. ACC is an accumulator for intermediate arithmetic results in this teaching model. A branch instruction may replace the normal next value of PC.

A useful analogy

Think of a recipe: PC marks the next recipe line, MAR selects its location, MDR carries the line, CIR holds the instruction being followed, and ACC is a temporary mixing bowl. In this simplified one-word instruction model PC advances by 1; actual instruction sizes and timing depend on the processor.

Apply and check

Try before reading feedback: While fetching the instruction at address 100, which register identifies the address and which carries the instruction from memory?

Feedback: MAR contains 100; MDR carries the fetched instruction. CIR holds the current instruction after the relevant transfer.

Practice connection: Exercise 1.3 in exercises.md; explained answer in solutions.md.

From source code to execution

Three translator roles

Tool	Input and result
Assembler	Assembly mnemonics and labels become machine instructions.
Compiler	Source program becomes another representation, often native code or bytecode.
Interpreter	Runs program operations through a runtime, sometimes using bytecode.

Choose the right level

Machine code contains the instructions a processor executes. Assembly exposes architecture-specific operations and registers. High-level languages offer abstractions such as variables, functions and data structures. They improve portability, but some translation or runtime support is still needed. A language may have both compiled and interpreted implementations.

Trade-offs are implementation-dependent

Ahead-of-time compilation can support fast execution and catch some errors before running. Interpretation can support rapid experimentation and interactive debugging. Many modern systems combine compilation, interpretation and just-in-time compilation. It is too simple to say that all interpreters translate one source line at a time or that all compilers produce a standalone executable.

Apply and check

Try before reading feedback: A language runs through bytecode in a virtual machine. Must it be called either compiled or interpreted, never both?

Feedback: No. A compiler may produce bytecode and a runtime may interpret or just-in-time compile it. Describe the implementation stages.

Practice connection: Exercise 1.5 in exercises.md; explained answer in solutions.md.

Trace a tiny program

Teaching assembly: one instruction per address

Address	Instruction	Meaning
100	LOAD #3	Put the literal value 3 in ACC.
101	ADD #4	Add the literal value 4 to ACC.
102	STORE 200	Copy ACC to memory address 200.
103	HALT	Stop this teaching program.

Watch state change

Start with PC = 100. After LOAD, ACC = 3. After ADD, ACC = 7. STORE copies 7 to address 200; it does not mean “store the number 200”. The # sign denotes an immediate value in this invented teaching notation. The example illustrates execution; it is not the instruction set of a particular real CPU.

Two passes and a helpful editor

A two-pass assembler first builds a table connecting labels to addresses, then uses it to resolve operands and encode instructions. An integrated development environment (IDE) combines editing, syntax highlighting, completion, translation and debugging. Breakpoints and single-stepping let you inspect the changing PC, registers and memory.

Apply and check

Try before reading feedback: Change ADD #4 to ADD #9 in the supplied program. Predict ACC and memory[200] after STORE.

Feedback: Both are 12: LOAD sets ACC to 3, ADD makes 12, and STORE copies the result. PC finally reaches 104 in this model after HALT is fetched.

Practice connection: Exercise 1.5 in exercises.md; explained answer in solutions.md.

The operating system's jobs

Coordinate shared resources

Management task	Purpose
Processes	Schedule CPU time and coordinate running programs.
Memory	Allocate and protect memory; manage virtual memory.
Devices and files	Use drivers, organise storage and provide I/O services.
Security	Authenticate users and enforce access permissions.

Utilities support maintenance

Backup software creates recoverable copies; test restoration as well as copying. Antivirus checks for known and suspicious malware activity. A formatter prepares a file system and may destroy access to existing data. Repair tools check file-system structures. Compression tools reduce storage or transfer size. Defragmentation helps some hard disks; it is not routine maintenance for SSDs.

Software does not replace hardware

The OS provides a managed interface to physical resources. A program still needs CPU time, working memory and access to its files. Good resource management improves sharing and responsiveness, but it does not remove limits in memory capacity, processing speed or storage performance.

Apply and check

Try before reading feedback: Pip loses its main drive after a backup was made yesterday. What should the team test before promising recovery?

Feedback: Restore the backup to a safe target and check required files and versions. A successful copy log alone does not establish recoverability.

Practice connection: Exercise 1.4 in exercises.md; explained answer in solutions.md.

Events, interrupts & peripherals

Respond without checking everything constantly

An interrupt requests the processor's attention. For an enabled interrupt, the processor and operating system preserve the required execution state, transfer control to an interrupt service routine, then restore state and resume. Exact timing and saved registers depend on the architecture. Not every request is handled immediately; priorities and masking affect when it is serviced.

Interrupt service, conceptually

Step / element	Meaning in the diagram
1	Running task
2	Save context
3	Service event
4	Restore / resume

A timer interrupt helps an OS share CPU time; an input device may interrupt when data becomes available.

Common peripheral connections

Connection	Typical purpose
USB	Serial data, plug-and-play peripherals; many versions also provide power.
HDMI	Digital video and audio for a display.
VGA	Legacy analogue video connection.

Apply and check

Try before reading feedback: A sensor interrupt arrives while a calculation is running. Why save context?

Feedback: The system needs enough state to resume the interrupted calculation correctly after the interrupt service routine.

Practice connection: Exercise 3.6 in exercises.md; explained answer in solutions.md.

Chapter review

Complete ACT-01 using the supplied mission instructions. Then choose two related terms and explain a mechanism using both; identify one limitation of the teaching model and one condition that could change the result.

Reference support

• R2: Nand to Tetris — Project 5: Computer Architecture — Project overview, proposed computer architecture and hardware simulator guidance. https://www.nand2tetris.org/project05

• R6: The Unicode Standard — technical introduction — Unicode design principles and character encoding model. https://www.unicode.org/standard/principles.html

Full resource introductions, optional tasks and access alternatives are in ../references.md.

Chapter 1 — Mission and optional resource

ACT-01 — The parcel total is wrong

Chapter 1 — The Fundamentals

Pip's delivery receipt says 200 parcels. The team expected 7. A programmer has confused an address with a value. Can you find the mistake?

Goal: Trace stored instructions and distinguish an immediate value, an address and the content stored there.

Preparation: Paper and pencil are sufficient. Optional: Python 3.9 or newer and supporting/BlueImpulse_Demonstrations.py. This is an invented teaching ISA, not a real processor.

Task:

1. Write columns for PC before fetch, instruction, ACC after execution and memory[200]. Start at PC 100, ACC 0 and memory[200] 0.

2. Run: 100 LOAD #3; 101 ADD #4; 102 STORE 200; 103 HALT. The # symbol means a literal value. PC advances by one when each instruction is fetched.

3. Predict the final result before checking the output. Then replace ADD #4 with ADD #9 and trace again.

4. Optional command: python3 supporting/BlueImpulse_Demonstrations.py --topic execution --add 9. Compare its state trace with your table.

Observe and record: Record ACC after every instruction, memory[200] after STORE, and final PC. Note the difference between a number used as data and a number used as an address.

Reflect: Why does STORE 200 copy 7 rather than replace the accumulator with 200? Which parts of the model would depend on a real ISA?

Feedback — check after trying: First run: after LOAD, ACC 3; after ADD, ACC 7; after STORE, memory[200] 7; after HALT, final PC 104. The modified run stores 12. STORE's operand is a destination address. PC increments, instruction size and register names are model choices, not universal CPU rules.

Optional resource: R2 (full introduction and offline alternative follow).

R2 — Nand to Tetris — Project 5: Computer Architecture

Noam Nisan and Shimon Schocken / Nand to Tetris.

https://www.nand2tetris.org/project05

Relevant section: Project overview, proposed computer architecture and hardware simulator guidance

Why use it: A concrete educational computer architecture connects logic, CPU state and memory.

Learner task: After ACT-01, compare the purpose of an instruction address and a data-memory address with the project's computer model. Name one part the models share and one detail that differs.

Limits: The Hack computer is a different ISA; its instructions and registers must not be substituted directly into this guide's toy program. Software installation is optional and not required for the paper task.

Offline alternative: ACT-01 includes its full program, trace and expected output. ACT-03 provides a complete paper pipeline grid specification.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

Chapter 1 — The Fundamentals

Key terms

Term	Meaning
Bit / byte	A binary digit / a group of eight bits.
Binary / denary / hexadecimal	Base-2 / base-10 / base-16 number systems.
Two's complement	A fixed-width signed representation with a negative most-significant-bit weight.
BCD	Four bits encode each individual decimal digit.
Character set / encoding	Available characters / how those characters are represented as data.
Resolution / colour depth	Pixel dimensions or density / bits used per pixel.
Sampling rate / bit depth	Samples per second / bits per sample.
Lossless / lossy	Exactly reversible / information-discarding compression.
Stored program	Instructions and data held in memory for execution.
ALU / control unit	Arithmetic and logic / coordination of operations.
PC / MAR / MDR	Next instruction address / accessed address / data being transferred.
CIR / ACC	Current instruction / intermediate arithmetic result in the teaching model.
Address / data / control bus	Location information / values and instructions / coordination signals.
Compiler / interpreter / assembler	Different mechanisms for translating or running programs.
Interrupt / context	A request for attention / the state needed to resume execution.
IDE	An integrated environment for editing, running and debugging code.

Chapter 1 — The Fundamentals

Exercises

Write reasoning, units and intermediate steps. All necessary concepts are explained in draft.md; the answer key is in solutions.md. The numbering preserves the source questions used throughout the guide.

Exercise 1.1a1

The byte 01001111 represents an eight-bit two's complement integer. State its equivalent denary value.

Exercise 1.1a2

The byte 10011010 represents an eight-bit two's complement integer. State its equivalent denary value.

Exercise 1.1a3

Write the integer −53 in eight-bit two's complement form.

Exercise 1.1a4

State the range of integers that can be represented by an eight-bit two's complement number. Give the endpoints in denary.

Exercise 1.1b1

Write the denary integer 798 in binary-coded decimal (BCD) format.

Exercise 1.1b2

Write the denary number represented by the BCD sequence 1001 0111 0111 0110.

Exercise 1.1c

Give one use of the binary-coded decimal system.

Exercise 1.2a1

A developer records sounds for a game. State what is meant by sampling resolution.

Exercise 1.2a2

Describe how sampling resolution affects the accuracy of stored digitised sound.

Exercise 1.2b1

A developer includes bitmap images in a game. Explain the term image resolution.

Exercise 1.2b2

A developer uses 16-colour bitmap images. State the number of bits required to encode the data for one pixel.

Exercise 1.2b3

An image is 16,384 pixels wide and 512 pixels high. It is saved as a 256-colour bitmap. Calculate the raw image-data size in gibibytes, excluding headers and compression.

Exercise 1.3

Put these fetch–execute stages in order: copy MDR to CIR; execute the instruction; decode the instruction; copy PC to MAR; increment PC by 1; copy the instruction at the memory location in MAR into MDR.

Exercise 1.4

File History and Time Machine are examples of backup utilities. Explain why it is important to back up files on a computer.

Exercise 1.5

Assemblers, compilers and interpreters are all used to translate or execute programs. Discuss the different roles played by each translator.

Chapter 1 — The Fundamentals

Explained answers

These are expected answers, not the only acceptable wording. Credit a correct alternative with clear assumptions and reasoning. Try the exercises before reading this section.

Answer 1.1a1

79. The sign bit is 0, so add the weights 64 + 8 + 4 + 2 + 1.

Answer 1.1a2

−102. In eight-bit two's complement, −128 + 16 + 8 + 2 = −102. Alternatively invert 10011010 and add 1 to obtain 01100110 = 102, then negate.

Answer 1.1a3

11001011. +53 is 00110101; invert to 11001010 and add 1. Check: −128 + 64 + 8 + 2 + 1 = −53.

Answer 1.1a4

−128 through +127 inclusive: −2^7 to 2^7 − 1. There are 256 distinct eight-bit patterns.

Answer 1.1b1

0111 1001 1000. Encode the decimal digits 7, 9 and 8 separately; do not convert the entire value to ordinary binary.

Answer 1.1b2

9776. The four nibbles represent decimal 9, 7, 7 and 6.

Answer 1.1c

A decimal digital display can store each displayed digit as one BCD nibble. Some decimal arithmetic systems also use BCD; it uses more bits than ordinary binary for most multi-digit values.

Answer 1.2a1

Sampling resolution (bit depth) is the number of bits used to encode the amplitude of each audio sample.

Answer 1.2a2

More bits per sample provide more quantisation levels, reducing amplitude quantisation error for a given signal range. This increases raw storage and does not itself increase the number of time samples per second.

Answer 1.2b1

For a stored bitmap, pixel dimensions describe the number of pixels across and down. In display or print contexts, resolution can also mean pixel density. State the intended meaning and units.

Answer 1.2b2

4 bits per pixel, because 2^4 = 16 colour codes.

Answer 1.2b3

256 colours require 8 bits = 1 byte per pixel. 16,384 × 512 = 8,388,608 bytes. Divide by 1,073,741,824 bytes/GiB to obtain 0.0078125 GiB (8 MiB). This excludes palette, header and compression.

Answer 1.3

PC → MAR; memory[MAR] → MDR; increment PC; MDR → CIR; decode; execute. This is the supplied one-word teaching model. Implementations can overlap operations and use different instruction lengths.

Answer 1.4

A backup permits recovery after accidental deletion, corruption, device failure or an attack. Keep a separate recoverable copy with appropriate access controls and test restoration; synchronisation alone can also synchronise a deletion.

Answer 1.5

An assembler translates assembly operations and labels into machine instructions. A compiler translates source into another representation, such as native instructions or bytecode. An interpreter executes operations through a runtime. Real implementations can combine these approaches; a language is not inherently limited to one.

Chapter 2 — Behaviorism & Behaviorist AI

Learning outcome: Distinguish classical and operant conditioning; identify stimulus–response rules and layered robotics; explain reinforcement-learning components, policies, values and exploration; interpret a Q update and the parts of an MDP.

Prerequisites: Read the relevant earlier chapters; this chapter connects those ideas to behaviorism & behaviorist ai.

Learning from the environment

Start with observable behaviour

Behaviourism studies the relationship between observable behaviour and environmental events. A stimulus is a detectable event; a response is an observable reaction. Pavlov, Watson and Skinner contributed to different strands of this tradition. The approach helped make learning measurable, although it is not a complete description of every kind of human learning.

Two kinds of conditioning

Classical conditioning	Operant conditioning
A cue becomes associated with another stimulus.	An action becomes associated with its consequences.
A bell paired with food can trigger salivation.	Sitting followed by a treat can become more frequent.
Focus: an elicited response.	Focus: behaviour shaped by reinforcement or punishment.

Connect the analogy carefully

Reinforcement learning borrows the idea of learning from consequences, but an algorithm is not a human or animal mind. A numerical reward is a training signal. It does not imply feelings, understanding or consciousness. Use the analogy to identify a feedback relationship, then describe the computational mechanism precisely.

Apply and check

Try before reading feedback: A robot receives reward +1 after turning left. Does that numerical signal prove it feels pleasure?

Feedback: No. Reward is an algorithmic training signal. The animal-learning analogy does not establish subjective experience.

Practice connection: Exercise 2.1 in exercises.md; explained answer in solutions.md.

React first: behaviour-based robots

A sensor can trigger a useful action

A simple robotic vacuum may move forward until a bumper is pressed, then reverse and turn. Its environment is the room, its stimulus is the bumper signal, and its response is the movement. These rules can produce useful behaviour without a detailed map. If the rules never change from experience, this is reactive control rather than reinforcement learning.

A direct feedback path

Step / element	Meaning in the diagram
1	Sensor
2	Behaviour rule
3	Actuator
4	Environment

Situated systems act in an environment; embodied systems have a physical body that constrains and enables action.

Layers and priorities

Brooks' subsumption architecture coordinates relatively simple behaviour modules. A module can inhibit or suppress another module's output under specified conditions. A designer might combine wandering, obstacle avoidance and task seeking while protecting a safety stop. This is one design approach, not a claim that all map-based planning is slow or unreliable; practical robots often combine reactive control and planning.

Apply and check

Try before reading feedback: Pip always reverses when its bumper triggers. No parameter ever changes. Is that alone evidence of reinforcement learning?

Feedback: No. It demonstrates a fixed reactive rule. RL requires a learning mechanism that uses experience to update values or a policy.

Practice connection: Exercise 2.3 in exercises.md; explained answer in solutions.md.

The reinforcement-learning loop

Learn a policy through experience

Reinforcement learning is a way for an agent to learn decisions through interaction. The agent observes a state or observation, selects an action, and receives feedback from the environment. A learning algorithm uses that experience to improve a policy: the rule or strategy for choosing actions. The objective is usually expected cumulative reward, not just the next reward.

One interaction at time t

Step / element	Meaning in the diagram
1	Agent / policy
2	Action
3	Environment
4	Next state + reward

Repeat: observe, act, receive feedback, then update. The learning update may occur during training rather than every deployed action.

Name the components

Component	Grid-world example
Agent / environment	A simulated rover / a grid with obstacles.
State / action	Current square / move up, down, left or right.
Reward / policy	Points for outcomes / a strategy for selecting moves.

Apply and check

Try before reading feedback: In a maze, identify an action and a reward without confusing them.

Feedback: "Move left" is an action; "−1 for that transition" is a numerical reward. The environment also returns a next state or observation.

Practice connection: Exercise 2.4a in exercises.md; explained answer in solutions.md.

Rewards, policies & values

Feedback evaluates an outcome

A reward is a numerical signal associated with an interaction. It may be positive, zero or negative. A reward of −1 is not a moral judgement: its role is defined by the task. Designers might reward reaching a goal, penalise a collision and charge a small cost per step. Poorly chosen rewards can encourage shortcuts that defeat the intended goal.

Distinguish the three ideas

The reward says how an outcome scores now. A policy says what action to choose in a situation. A value estimate says how much future reward is expected from a state or a state–action choice. A good policy may accept a small immediate cost to achieve a larger long-term return. Q-learning estimates action values and derives choices from them.

The learning question

Step / element	Meaning in the diagram
1	Experience
2	Value estimate
3	Policy improvement

A fixed rule such as "if hot, turn on fan" may control a system effectively without learning any values or changing a policy.

Apply and check

Try before reading feedback: A route costs −1 now but leads to a large later reward. Why might a good policy choose it?

Feedback: It can maximise expected cumulative return. Immediate reward and long-term value are different quantities.

Practice connection: Exercise 5.3a in exercises.md; explained answer in solutions.md.

Explore or exploit?

Use the restaurant analogy

Exploitation is ordering a dish you already believe is best. Exploration is trying another dish to learn more about it. In an RL task, exploitation chooses the action with the best current value estimate, including future return. Exploration gathers information about other actions. Early estimates can be wrong, so always exploiting can lock an agent into an inferior policy.

An epsilon-greedy example

Choice rule	With epsilon = 0.10
Explore	With 10% probability, choose a random permitted action.
Exploit	Otherwise choose an action with the highest current Q value.
During training	Measure outcomes and update estimates; epsilon may change over time.

Think before choosing

Suppose one route has worked twice and another has never been tried. Is the first route certainly better? No: evidence is incomplete. Exploration can improve knowledge, but it also has a cost. In physical systems, experiments must respect safety constraints; training in a simulation does not automatically make every real-world action safe.

Apply and check

Try before reading feedback: A policy always selects the route with the best estimate from just one trial. What is the risk?

Feedback: An inaccurate early estimate may never be corrected because other routes are not explored. Exploration provides evidence but may have costs.

Practice connection: Exercise 2.4b in exercises.md; explained answer in solutions.md.

Update a Q value

A table of learned action values

Q(s, a) estimates the return from taking action a in state s and then following a good policy. Q-learning is model-free: it learns from transitions without needing a supplied transition model. "Model-free" does not mean memory-free or that the agent lacks a table or neural network.

The update rule

Q(new) = Q(old) + alpha × [reward + gamma × max Q(next state, next action) − Q(old)]. Alpha controls the size of an update. Gamma discounts future rewards. The quantity in square brackets is the temporal-difference error. For a terminal next state, use zero for the future-value term.

Work through one illustrative update

Quantity	Value
Old Q / alpha	2 / 0.5
Reward / gamma / best next Q	1 / 0.9 / 4
Target	1 + 0.9 × 4 = 4.6
New Q	2 + 0.5 × (4.6 − 2) = 3.3

Interpret the result

This experience raised the estimate from 2 to 3.3. It did not prove the action is always best. Repeated experience, an appropriate learning schedule and sufficient exploration are important; one reward does not establish a reliable policy.

Apply and check

Try before reading feedback: For old Q = 2, alpha = 0.5 and a terminal reward of 1, what is new Q?

Feedback: At a terminal state future value is zero: 2 + 0.5 × (1 − 2) = 1.5.

Practice connection: Exercise 2.4b in exercises.md; explained answer in solutions.md.

A decision problem as an MDP

Five parts of a Markov decision process

Symbol	Meaning
S	Possible states.
A	Available actions.
P	Transition probabilities: where an action can lead.
R	Reward function: how outcomes are scored.
gamma	Discount factor: how future rewards are weighted.

The Markov property

A state is Markov when it contains the information needed to predict the next-state and reward distribution, given the current action. Earlier history adds no further predictive information once that state is known. A position alone may be insufficient: a robot may also need speed, battery level or whether it is carrying a package.

Observation is not always state

Sensors can provide incomplete or noisy observations. A camera view may hide an obstacle or omit velocity. An agent may need memory or a state estimate to work with partial information. A simple grid world is useful because we can explicitly specify the state, actions, transitions and rewards instead of assuming the agent knows everything.

Apply and check

Try before reading feedback: Two robots occupy the same square but have different speeds. Can position alone always be a Markov state?

Feedback: No. If speed changes the distribution of next outcomes, a sufficient state must include it or an equivalent representation.

Practice connection: Exercise 2.4a in exercises.md; explained answer in solutions.md.

Actions can have uncertain outcomes

A slippery grid square

Imagine an agent choosing “forward”. In this illustrative environment, it reaches the intended square with probability 0.8 and slips sideways with probability 0.2. The action is chosen by the agent; the transition outcome is sampled by the environment. The probabilities for all possible next outcomes must sum to 1.

One action, two outcomes

Step / element	Meaning in the diagram
1	Choose forward
2	Target: probability 0.8
3	Slip: probability 0.2

For immediate rewards of +10 and −5, the expected immediate reward is 0.8 × 10 + 0.2 × (−5) = 7. Future rewards may change which action is best.

Expected is not guaranteed

An expected value of 7 does not mean every attempt earns 7. It summarises the average implied by the model. A system that learns under one set of transition probabilities may perform poorly when its environment changes. Test across plausible conditions rather than judging a policy from a single successful run.

Apply and check

Try before reading feedback: Pip earns +10 with probability 0.8 and −5 with probability 0.2. Must one trial earn the expected value 7?

Feedback: No. The expectation is 0.8 × 10 − 0.2 × 5 = 7, but an individual trial has one of the defined outcomes.

Practice connection: Exercise 2.4a in exercises.md; explained answer in solutions.md.

Training is different from using a policy

Separate two phases

During training, the system collects experience and updates estimates or policy parameters. During deployment, it may use a fixed learned policy without further learning. Some applications allow controlled online updates, but adaptation is not automatic. A robot can react to a new sensor reading while keeping exactly the same policy.

Test what was actually learned

Check whether success persists with different starting states, sensor noise and changed conditions. Compare against a simple fixed-rule baseline. Examine failure cases as well as average reward. If the goal is “arrive safely”, a reward that only measures speed may produce an undesirable strategy even when the algorithm is working as specified.

A responsible learning workflow

Step / element	Meaning in the diagram
1	Define the task
2	Train / explore
3	Evaluate
4	Deploy / monitor

Behaviour-based robotics, reinforcement learning and neural networks are related ideas, not interchangeable labels. A system can use one without using all three.

Apply and check

Try before reading feedback: Pip reacts to a newly seen obstacle using a frozen trained policy. Did that reaction necessarily train the model?

Feedback: No. Inference can choose a different action from a new observation without changing any learned parameter.

Practice connection: Exercise 2.2 in exercises.md; explained answer in solutions.md.

Chapter review

Complete ACT-02 using the supplied mission instructions. Then choose two related terms and explain a mechanism using both; identify one limitation of the teaching model and one condition that could change the result.

Reference support

• R1: Reinforcement Learning: An Introduction, second edition — Chapter 3 (finite Markov decision processes) and Section 6.5 (Q-learning). http://incompleteideas.net/book/the-book-2nd.html

Full resource introductions, optional tasks and access alternatives are in ../references.md.

Chapter 2 — Mission and optional resource

ACT-02 — One lucky route is not a policy

Chapter 2 — Behaviorism & Behaviorist AI

Pip takes a new route and arrives early once. "Best route forever!" it announces. Its teammate asks whether one reward is enough evidence.

Goal: Calculate a Q-learning update and distinguish an estimate from a guaranteed outcome.

Preparation: Calculator or paper. Optional standard-library Python script. Know reward, policy, value, alpha and gamma from this chapter.

Task:

1. Use old Q = 2, alpha = 0.5, reward = 1, gamma = 0.9 and best next Q = 4. Predict whether the estimate will rise or fall.

2. Calculate target = reward + gamma × best next Q; error = target − old Q; change = alpha × error; new Q = old Q + change.

3. Repeat with the same old Q, alpha and reward but a terminal next state. Set future value to zero.

4. Optional command: python3 supporting/BlueImpulse_Demonstrations.py --topic rl. Compare values. Explain whether either update proves a route is safest.

Observe and record: Record the target, temporal-difference error, change and new value for both cases.

Reflect: Why is the future term zero at termination? Why can exploration remain useful after a positive outcome? What safety rule would constrain a real robot's exploration?

Feedback — check after trying: Non-terminal: target 4.6, error 2.6, change 1.3, new Q 3.3. Terminal: target 1, error −1, change −0.5, new Q 1.5. These are updated estimates, not proofs. Use many relevant transitions and evaluate behaviour against the task, including safety constraints.

Optional resource: R1 (full introduction and offline alternative follow).

R1 — Reinforcement Learning: An Introduction, second edition

Richard S. Sutton and Andrew G. Barto, MIT Press, 2018.

http://incompleteideas.net/book/the-book-2nd.html

Relevant section: Chapter 3 (finite Markov decision processes) and Section 6.5 (Q-learning)

Why use it: Primary textbook support for the agent/environment distinction, expected return and the Q-learning update.

Learner task: After ACT-02, explain in your own words what the bootstrapped target includes, then compare it with the terminal-state case.

Limits: The book is mathematically deeper than this beginner guide. Start with the supplied worked calculation; do not treat the browser page as required activity content.

Offline alternative: The formula, both complete calculations and interpretation are included in ACT-02 and the chapter, so the task works offline.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

Chapter 2 — Behaviorism & Behaviorist AI

Key terms

Term	Meaning
Behaviourism	An approach studying observable behaviour and its relation to the environment.
Stimulus / response	A detectable event / the observable reaction.
Subsumption	Layered behaviour coordination using suppression or inhibition.
Agent / environment	The decision-maker / the system it interacts with.
State / observation	A representation sufficient for a task / the information actually sensed.
Action / reward	A choice / a numerical feedback signal.
Policy	A strategy that maps a situation to action choices.
Value / Q value	Expected return from a state / from a state–action choice.
Exploration / exploitation	Gathering new experience / using current value estimates.
Learning rate	A parameter controlling the size of an update.
Discount factor	A parameter controlling the weighting of future rewards.
MDP	A framework specifying states, actions, transitions, rewards and discounting.
Model-free RL	Learning without a supplied environment transition model.
Markov property	The current state and action contain the information needed for the next-step distribution.

Chapter 2 — Behaviorism & Behaviorist AI

Exercises

Write reasoning, units and intermediate steps. All necessary concepts are explained in draft.md; the answer key is in solutions.md. The numbering preserves the source questions used throughout the guide.

Exercise 2.1

Differentiate between classical conditioning (Pavlov) and operant conditioning (Skinner) in terms of how learning occurs.

Exercise 2.2

Give an example of a simple reflex-based agent, such as a robotic vacuum or thermostat, and identify its stimulus, response and environment.

Exercise 2.3

Explain how subsumption architecture organises robot behaviours into layered modules without requiring a centralised internal map.

Exercise 2.4a

What are the roles of the environment, the agent and the policy in a Markov decision process (MDP)?

Exercise 2.4b

Explain the difference between exploration (trying new actions) and exploitation (using the best current estimates). Why must an agent balance both?

Chapter 2 — Behaviorism & Behaviorist AI

Explained answers

These are expected answers, not the only acceptable wording. Credit a correct alternative with clear assumptions and reasoning. Try the exercises before reading this section.

Answer 2.1

Classical conditioning associates a cue with another stimulus so the cue can elicit a response. Operant conditioning changes the likelihood of an action through its consequences. A bell paired with food and a voluntary action followed by a reward illustrate the difference. They are analogies, not proof that numerical RL reproduces animal experience.

Answer 2.2

A thermostat is a fixed reflex example: environment = room; stimulus = measured temperature below the set point; response = switch heating on. With unchanged thresholds it reacts but does not learn a policy from rewards.

Answer 2.3

Subsumption coordinates behaviour modules with defined priority, inhibition or suppression relationships. Obstacle avoidance may suppress a wandering command. Useful behaviour can emerge without a single detailed central map, although practical robots can combine reactive modules with mapping and planning.

Answer 2.4a

The agent chooses an action using its policy; the environment produces the next state and reward according to the task's dynamics. An MDP specifies states, actions, transition probabilities, rewards and discounting. A policy maps states to actions or action probabilities; it is not the reward function.

Answer 2.4b

Exploration obtains information about insufficiently tested actions. Exploitation chooses an action using the best current value estimate, which may include future rewards. Always exploiting can lock in an inferior early estimate; always exploring can waste known good actions. Physical exploration needs safety constraints.

Chapter 3 — Hardware

Learning outcome: Classify hardware; compare memory and storage; explain processors, pipelining, parallelism and interrupts; connect sensors, converters and actuators; reason with logic gates; evaluate embedded systems, interfaces and AI accelerators.

Prerequisites: Read the relevant earlier chapters; this chapter connects those ideas to hardware.

Hardware gives software a body

Classify by the job being performed

Hardware is the physical part of a computer system. An input device supplies data, a processing device transforms it, an output device communicates or acts on a result, and storage retains information. One device can have several roles: a touchscreen displays output and senses input. A barcode reader is input; an SSD is storage.

From a physical event to a response

Step / element	Meaning in the diagram
1	Sensor / input
2	Processor + memory
3	Output / actuator

The processor follows instructions. Useful behaviour depends on the program, the data, the hardware and their connections.

How hardware enables AI

An AI model must run as computations on hardware. Sensors provide observations, memory holds model parameters and working data, a processor evaluates the model, and an output device communicates the decision. A physical robot also needs a powered actuator to move. More powerful hardware can increase capacity or speed, but does not itself create a learning algorithm.

Apply and check

Try before reading feedback: Classify a touch panel, a camera and a wheel motor by their role in Pip.

Feedback: Touch panel: input and output; camera: input; wheel motor: actuator/output. Roles describe what devices do, not just their shape.

Practice connection: Exercise 3.3 in exercises.md; explained answer in solutions.md.

Memory: capacity versus speed

A useful hierarchy

Resource	Role and trade-off
Registers	Very small, very fast storage inside the processor.
Cache	Fast memory holding copies of recently or frequently used data.
Main RAM	Working storage for active programs and data; normally volatile.
Secondary storage	Larger, persistent storage; usually slower to access.

SRAM and DRAM

Static RAM stores bits using bistable circuits such as flip-flops. It needs power but no periodic refresh, and is commonly used for CPU caches. Dynamic RAM stores charge in cells that must be refreshed; its greater density and lower cost make it suitable for main memory. Both are volatile. Cache can be on the processor chip; it is not necessarily external.

Why an AI workload needs memory

Model weights, input data and intermediate results must fit somewhere. A fast processor can spend time waiting for data if memory bandwidth is insufficient. Capacity answers “how much fits?” Bandwidth answers “how quickly can it move?” These are different constraints and both affect performance.

Apply and check

Try before reading feedback: A model’s weights fit SSD storage but not working RAM. Why can execution still fail?

Feedback: Storage capacity does not provide the required fast working memory. Weights, activations and runtime data must fit the execution strategy’s memory budget.

Practice connection: Exercise 3.2a in exercises.md; explained answer in solutions.md.

Keep data when power is off

ROM and flash families

Type	Writing or erasing
Mask ROM	Programmed during manufacture.
PROM	Programmed once after manufacture.
EPROM	Erased with ultraviolet light, then reprogrammed.
EEPROM	Electrically erasable and reprogrammable.
Flash	Electrically erasable memory, normally erased in blocks.

Magnetic, solid-state and optical media

A hard disk stores data magnetically on rotating platters. Seek time and rotational latency delay access; fragmentation can increase head movement. An SSD stores data in flash with no moving parts. It avoids mechanical seek and rotation delays, but still has nonzero access latency and finite write endurance. External drives can provide portable backup storage.

Optical storage

CDs, DVDs and Blu-ray discs use laser light to read recorded features. Some discs have multiple recording layers. Material defects and optical effects, including birefringence, can disrupt reading. Choose a medium using capacity, access time, cost, durability and the need for long-term retention rather than assuming one technology is best for every purpose.

Apply and check

Try before reading feedback: Which is normally changed electronically: EPROM or EEPROM?

Feedback: EEPROM is electrically erased and rewritten; EPROM traditionally needs ultraviolet erasure. Both are non-volatile.

Practice connection: Exercise 3.2b in exercises.md; explained answer in solutions.md.

Processor instruction-set choices

Introductory design tendencies

RISC	CISC
Regular, often simpler instruction encodings.	Can include more complex and variable-length instructions.
Typically separates loads/stores from register arithmetic.	May allow operations with memory operands.
Examples include ARM and RISC-V families.	A major example is the x86 family.

The distinction is not a speed guarantee

A reduced instruction set does not guarantee that every instruction takes one clock cycle. A complex instruction set does not prevent pipelining or high performance. Modern processors use caches, pipelines, prediction and other techniques; many translate instructions internally into simpler operations. Some RISC families also support compressed instruction encodings.

Look at the workload

Performance depends on implementation, clock frequency, instructions per cycle, memory behaviour, software and energy constraints. A mobile device may prioritise power efficiency; a workstation may prioritise throughput. Compare measured work done under relevant conditions instead of judging a CPU from the RISC or CISC label alone.

Apply and check

Try before reading feedback: A seller claims "RISC" proves a chip is faster. Name two measurements you still need.

Feedback: Measure the relevant workload's time/throughput and energy under stated conditions. Memory, microarchitecture, compiler and clock also matter.

Practice connection: Exercise 3.5a in exercises.md; explained answer in solutions.md.

Overlap work with a pipeline

Use the laundry analogy

A laundry can wash one load while drying an earlier load. Similarly, a processor pipeline can fetch a new instruction while decoding or executing previous ones. This increases throughput after the pipeline fills; it does not necessarily reduce the latency of an individual instruction. Pipelining is used in both RISC and CISC implementations.

A simplified four-stage pipeline

Instruction	Cycle 1	2	3	4	5
I1	Fetch	Decode	Execute	Write	—
I2	—	Fetch	Decode	Execute	Write
I3	—	—	Fetch	Decode	Execute

Why a pipeline can pause

A data hazard occurs when an instruction needs a result that is not ready. A control hazard occurs when a branch makes the next instruction uncertain. A structural hazard occurs when operations need the same resource. Stalling, forwarding, scheduling and prediction can help, depending on the processor. "Overlapping" does not mean every instruction is independent.

Apply and check

Try before reading feedback: Four equal pipeline stages process five independent instructions without stalls. How many cycles?

Feedback: $4 + 5 - 1 = 8$ cycles. This ideal calculation concerns total throughput; one instruction still needs four stage cycles.

Practice connection: Exercise 3.5b in exercises.md; explained answer in solutions.md.

One stream or many?

Flynn's classification

Class	Instruction / data streams	Example idea
SISD	Single / single	One instruction stream processes individual data.
SIMD	Single / multiple	One operation applied to many array elements.
MISD	Multiple / single	Rare; different processing of one data stream.
MIMD	Multiple / multiple	Independent cores or computers run different work.

Scale beyond one processor

A cluster connects computers, called nodes, so they can coordinate work. Massively parallel systems distribute large tasks across many processing units. Supercomputers are designed for demanding scientific or engineering computation; they are not simply a synonym for mainframes. Communication, synchronisation and data movement affect whether adding processors actually helps.

Connect this to AI

Large arrays of multiplications can benefit from parallel hardware. Parts of an AI workload may still be sequential or limited by memory and communication. A GPU has its own execution model, often described using SIMT, and provides SIMD-like parallel arithmetic; the simple taxonomy is a starting point, not a complete description of every chip.

Apply and check

Try before reading feedback: One operation is applied to many image pixels at once. Which Flynn category is the clearest introductory match?

Feedback: SIMD: one instruction stream operating on multiple data elements. Real GPUs use further scheduling details beyond this simplified classification.

Practice connection: Exercise 3.5b in exercises.md; explained answer in solutions.md.

Let hardware request attention

Events can change the schedule

A sensor, network interface or timer can raise an interrupt. The CPU responds when the request is enabled and eligible. It preserves enough context to resume the interrupted work and transfers execution to a handler. The handler deals with the event and returns. The hardware and operating system divide the responsibility for saving and restoring state.

A simplified service sequence

Step / element	Meaning in the diagram
1	Event arrives
2	Preserve context
3	Run handler
4	Resume task

Context can include the program counter, status and required registers. Exact details vary by architecture.

Predictability matters

A system with a timing deadline must account for interrupt priority, handler execution time and other sources of delay. RISC versus CISC alone does not determine interrupt latency. Long or frequent interrupt handling can delay useful work. A well-designed system measures worst-case behaviour rather than assuming every event will receive an immediate response.

Apply and check

Try before reading feedback: Why not promise that every interrupt is serviced instantly?

Feedback: Interrupts can be masked or delayed by higher-priority work; architecture and system policy determine response time.

Practice connection: Exercise 3.6 in exercises.md; explained answer in solutions.md.

Reason with logic gates

Gates transform binary signals

AND produces 1 only when all inputs are 1. OR produces 1 when at least one input is 1. NOT inverts one input. XOR produces 1 when two inputs differ. NAND is the inverse of AND; NOR is the inverse of OR. A truth table lists every possible input combination and its corresponding output.

Two-input truth table

A	B	AND	OR	XOR	NAND	NOR
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	1	0
1	1	1	1	0	0	0

Read symbols and conditions

NOT maps 0 to 1 and 1 to 0. An AND symbol has a flat input side and curved output side; an OR symbol has a curved input side and pointed output. A small output circle means inversion. In a control rule, “door closed AND start pressed” requires both conditions; “smoke OR manual alarm” requires either.

Apply and check

Try before reading feedback: A rover moves only when battery_ok AND NOT obstacle. Evaluate battery_ok = 1 and obstacle = 1.

Feedback: NOT obstacle = 0, so 1 AND 0 = 0: do not move.

Practice connection: Exercise 3.7 in exercises.md; explained answer in solutions.md.

Simplify a logic expression

Useful identities

Law	Equivalent expressions
Identity	A AND 1 = A; A OR 0 = A
Null	A AND 0 = 0; A OR 1 = 1
Idempotent	A AND A = A; A OR A = A
Complement	A AND NOT A = 0; A OR NOT A = 1
De Morgan	NOT(A AND B) = (NOT A) OR (NOT B)
De Morgan	NOT(A OR B) = (NOT A) AND (NOT B)

A worked simplification

Let X = (A AND B) OR (A AND NOT B). Factor out A: X = A AND (B OR NOT B). The term in brackets is always 1, so X = A AND 1 = A. A simpler equivalent circuit can reduce hardware and delay. Check equivalence by comparing truth-table outputs for every input combination.

Keep the negation in the right place

De Morgan's laws negate the whole grouped expression on the left. Dropping that outer NOT changes the meaning. Parentheses make the scope clear, especially when a printed overbar is difficult to see. In Boolean algebra, OR is not ordinary arithmetic addition: 1 OR 1 is still 1.

Apply and check

Try before reading feedback: Test whether A OR (A AND B) equals A when A = 0, B = 1. Then test A = 1, B = 0.

Feedback: First: 0 OR (0 AND 1) = 0. Second: 1 OR (1 AND 0) = 1. Both match A; a full four-row truth table confirms absorption.

Practice connection: Exercise 3.7 in exercises.md; explained answer in solutions.md.

Sense, decide, act, sense again

Monitoring versus control

A monitoring system records or displays measurements. A control system also acts on the environment. In a greenhouse, a temperature sensor supplies a reading, a processor compares it with a target, and a fan changes the temperature. A later measurement closes the feedback loop. This can be a fixed-rule controller; learning is not required.

An environmental control loop

Step / element	Meaning in the diagram
1	Sensor
2	Input conversion
3	Decision
4	Driver + actuator

The changed environment is measured again. Analogue sensors may need an ADC; many sensors already provide digital data.

Signals are not actuator power

An ADC converts an analogue signal to a digital value. A DAC converts a digital value to an analogue signal. Some actuators use PWM or digital interfaces instead of a DAC. PWM is a timed switching signal, not the same thing as analogue conversion. A motor driver or power stage supplies the current needed to move a motor or operate a valve.

Apply and check

Try before reading feedback: A fan switches on above 30°C and off below 28°C. Why use two thresholds?

Feedback: The gap provides hysteresis so tiny fluctuations near one threshold do not repeatedly switch the actuator on and off.

Practice connection: Exercise 3.4 in exercises.md; explained answer in solutions.md.

Computing inside another device

A dedicated purpose

An embedded system performs a particular role within a larger product, such as a washing-machine controller or a sensor node. A microcontroller typically combines a CPU, memory and input/output peripherals on one chip. An embedded system may use a microcontroller, a larger processor or specialised hardware, depending on its requirements.

Benefits and constraints

Potential benefit	Constraint to consider
Compact, low-power operation.	Limited memory, processing capacity and battery energy.
Low cost for repeated tasks.	Upgrades and repairs may be difficult.
Predictable behaviour with careful design.	Timing and reliability must be tested, not assumed.
Local operation without a cloud link.	External data or remote management may still need a network.

Design around the task

A small fixed-rule controller may need only modest hardware. A vision model can require much more memory and computation. Choose sensors, processing capacity, interfaces and power together. Environmental temperature, moisture, vibration and expected service life also influence whether a design is suitable.

Apply and check

Try before reading feedback: Why may a campus rover use a microcontroller for a motor loop even if it also has a large CPU?

Feedback: A dedicated embedded controller can service a narrow timing-critical job with appropriate I/O and power requirements. The exact choice still needs measurements.

Practice connection: Exercise 6.1 in exercises.md; explained answer in solutions.md.

The hardware behind an AI model

Different processing strengths

Hardware	Useful role
CPU	General control, branching and diverse program tasks.
GPU	High-throughput parallel arithmetic for suitable workloads.
NPU / TPU-type accelerator	Specialised tensor operations with hardware-specific efficiency.
RAM and storage	Hold weights, inputs, intermediate results and saved models.

Training and inference have different costs

Training adjusts model parameters using data or experience. Inference uses a model to produce an output. A model may be trained on a large server and deployed on a smaller device. Simple tabular reinforcement learning can run on an ordinary CPU; AI does not always require a GPU, and a specialised chip is not automatically faster for every workload.

Trace the whole path

For a camera-based robot, the sensor captures an image, memory buffers it, processing estimates a useful state, a policy selects an action, and control hardware drives the actuator. Delays in any stage matter. Model size, memory movement, power consumption and heat can be as important as the advertised arithmetic speed of the chip.

Apply and check

Try before reading feedback: What is more naturally parallel: multiplying a large matrix or following a dependent chain of branch decisions?

Feedback: The matrix often exposes many independent arithmetic operations. The branch chain can have sequential dependencies, so a GPU is not universally fastest.

Practice connection: Exercise 6.1 in exercises.md; explained answer in solutions.md.

Interfaces you can see and touch

Touch and display technologies

Technology	Key idea
Capacitive touch	Detects changes in an electric field; often supports multi-touch.
Resistive touch	Pressure brings conductive layers into contact.
OLED display	Organic light-emitting pixels produce their own light.
VR headset	Combines displays, optics and tracking to present an immersive view.

Display quality involves trade-offs

Screen resolution describes pixel dimensions. OLED can provide deep blacks because individual pixels can be off, but power use depends on displayed content and hardware. Capacitive and resistive touch differ in input methods, clarity, durability and cost. A touchscreen is both an input surface and an output display.

Build physical objects layer by layer

In material-extrusion 3D printing, a nozzle deposits material to form successive layers. In binder-based printing, a binder joins selected regions of a powder bed. These are different additive-manufacturing methods; material properties and post-processing requirements differ. A printer turns digital instructions into physical output through controlled motion and material handling.

Apply and check

Try before reading feedback: A voice interface succeeds only in quiet rooms. What should a usable rover interface add?

Feedback: An accessible alternative such as a clear button or text/touch interface, visible feedback and a recovery path. Test actual users and noise conditions.

Practice connection: Exercise 3.3 in exercises.md; explained answer in solutions.md.

Chapter review

Complete ACT-03 using the supplied mission instructions. Then choose two related terms and explain a mechanism using both; identify one limitation of the teaching model and one condition that could change the result.

Reference support

• R2: Nand to Tetris — Project 5: Computer Architecture — Project overview, proposed computer architecture and hardware simulator guidance. https://www.nand2tetris.org/project05

Full resource introductions, optional tasks and access alternatives are in ../references.md.

Chapter 3 — Mission and optional resource

ACT-03 — The kitchen with four stations

Chapter 3 — Hardware

Pip needs three sensor calculations. One worker finishes all four steps of a job before starting the next. Another team proposes four stations. Will each job suddenly take one step?

Goal: Distinguish pipeline throughput from the latency of one instruction and identify the effect of a dependency.

Preparation: Draw a grid with cycles 1–8 as columns and instructions I1–I3 as rows. Use stages F (fetch), D (decode), E (execute), W (write back). Assume one cycle per stage, independent instructions and no stalls initially.

Task:

1. Schedule I1 in F/D/E/W across cycles 1–4; start I2 at cycle 2 and I3 at cycle 3.

2. Predict the finish cycles and compare total elapsed cycles with three instructions run without overlap.

3. Now assume I2 needs a result from I1 that is unavailable until the end of I1's W. State this dependency explicitly; postpone I2's E to cycle 5 and propagate later stages without letting instructions overtake.

4. Optional command: python3 supporting/BlueImpulse_Demonstrations.py --topic hardware. Use its ideal trace for the first case. Treat your stalled case as a separate scheduling model.

Observe and record: Ideal finish cycles are 4, 5 and 6. Mark every occupied stage and every inserted wait. Do not equate elapsed cycles with the sum of all stage work.

Reflect: What improved: single-instruction latency or completed instructions per unit time? Why must a claim of speedup list its assumptions?

Feedback — check after trying: Ideal case takes six cycles rather than twelve sequential stage cycles, but each instruction still uses four stages. In the stated dependency model I2 executes at 5 and writes at 6; I3 must wait behind it and can execute at 6 and write at 7. Real forwarding, stage timings and hazards differ, so this trace does not predict every CPU.

Optional resource: R2 (full introduction and offline alternative follow).

R2 — Nand to Tetris — Project 5: Computer Architecture

Noam Nisan and Shimon Schocken / Nand to Tetris.

https://www.nand2tetris.org/project05

Relevant section: Project overview, proposed computer architecture and hardware simulator guidance

Why use it: A concrete educational computer architecture connects logic, CPU state and memory.

Learner task: After ACT-01, compare the purpose of an instruction address and a data-memory address with the project's computer model. Name one part the models share and one detail that differs.

Limits: The Hack computer is a different ISA; its instructions and registers must not be substituted directly into this guide's toy program. Software installation is optional and not required for the paper task.

Offline alternative: ACT-01 includes its full program, trace and expected output. ACT-03 provides a complete paper pipeline grid specification.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

Chapter 3 — Hardware

Key terms

Term	Meaning
Hardware	Physical computing components.
Input / output / storage	Data entry / results or action / retention of information.
Volatile / non-volatile	Loses data without power / retains data without power.
SRAM / DRAM	Bistable-cell memory / charge-based memory requiring refresh.
ROM / PROM / EPROM	Read-only / one-time programmable / ultraviolet-erasable memory.
EEPROM / flash	Electrically erasable memory / a block-erasable family.
Cache	Fast storage holding copies of data near a processor.
HDD / SSD / optical	Magnetic disk / solid-state storage / laser-readable media.
RISC / CISC	Instruction-set design families; not direct performance guarantees.
Pipeline / hazard	Overlapped instruction stages / a dependency or conflict that can stall them.
SISD / SIMD / MISD / MIMD	Flynn's categories of instruction and data streams.
Sensor / actuator	Measures a physical condition / produces physical action.
ADC / DAC / PWM	Analogue-to-digital conversion / digital-to-analogue conversion / pulse-width modulation.
Microcontroller	An integrated processor with memory and input/output peripherals.
Embedded system	A computing system with a role inside a larger product.
GPU / NPU	Parallel graphics-oriented processor / specialised neural-processing hardware.
Logic gate / truth table	A binary operation circuit / its outputs for every input combination.

Chapter 3 — Hardware

Exercises

Write reasoning, units and intermediate steps. All necessary concepts are explained in draft.md; the answer key is in solutions.md. The numbering preserves the source questions used throughout the guide.

Exercise 3.1a

Differentiate between primary storage and secondary storage in terms of volatility, speed and CPU accessibility.

Exercise 3.1b

Give an example of a secondary-storage technology for each category: magnetic, optical and solid-state (flash).

Exercise 3.2a

Explain the structural difference between SRAM and DRAM regarding how they store data and their need for refreshing.

Exercise 3.2b

Contrast ROM, PROM and EPROM in terms of how data is written and erased.

Exercise 3.3

Classify the following as input, output or storage, noting dual roles where relevant: a touchscreen display, a solid-state drive and a barcode reader.

Exercise 3.4

Outline an automated environmental control system, such as a greenhouse or thermostat. Specify the sensor, the processing condition and the resulting actuator action.

Exercise 3.5a

Explain the main design difference between RISC and CISC architectures regarding instruction encoding and decoding complexity.

Exercise 3.5b

Explain the role of pipelining in overlapping instruction execution, and name one hazard that can stall a pipeline.

Exercise 3.5c

State the functions of the program counter, accumulator and memory data register during instruction execution.

Exercise 3.6

What is a hardware interrupt, and what happens to program execution when an enabled interrupt is serviced by the CPU?

Exercise 3.7

Draw or describe the logic-gate symbols and write the truth tables for AND, OR and NOT.

Chapter 3 — Hardware

Explained answers

These are expected answers, not the only acceptable wording. Credit a correct alternative with clear assumptions and reasoning. Try the exercises before reading this section.

Answer 3.1a

Primary storage includes working memory and, in some classifications, ROM; it is accessed through the processor's memory system. Main RAM is normally volatile and relatively fast. Secondary storage such as SSD or HDD is non-volatile and usually slower to access. Data normally moves into working memory before ordinary processing. "All primary storage is volatile" is false.

Answer 3.1b

Magnetic: hard disk drive or magnetic tape. Optical: CD, DVD or Blu-ray. Solid-state: SSD or USB flash drive.

Answer 3.2a

SRAM uses a bistable transistor cell and does not need periodic refresh while powered. DRAM stores charge that leaks and needs refresh. Both normally lose data without power. SRAM is fast but less dense and costlier per bit; DRAM supports larger main memories.

Answer 3.2b

Mask ROM is fixed during manufacture. PROM can normally be programmed once. EPROM can be erased with ultraviolet light and reprogrammed; EEPROM and flash instead use electrical erasure. These names refer to different technologies, not one universal write process.

Answer 3.3

Touchscreen = output display plus touch input. SSD = storage, with read/write data transfers. Barcode reader = input. A device can legitimately have more than one role.

Answer 3.4

For a greenhouse, a temperature sensor provides readings to a controller; when the temperature exceeds a defined threshold, the controller drives a fan actuator. Measure again to close the loop. A lower switch-off threshold can prevent rapid on/off chatter. This is fixed-rule control unless a learning method actually updates the decision policy.

Answer 3.5a

RISC families traditionally emphasise relatively regular, simple instruction encodings and load/store operations. CISC families traditionally provide more varied encodings and operations. Modern implementations blur boundaries and may translate complex instructions into micro-operations. Neither label alone proves speed or efficiency.

Answer 3.5b

Pipelining overlaps stages of different instructions to increase throughput. In an ideal four-stage pipeline, three independent instructions need 4 + 3 − 1 = 6 cycles, rather than 12 sequential stage cycles. A data dependency, resource conflict or branch can cause stalls; one instruction still traverses all four stages.

Answer 3.5c

PC holds the next instruction address; ACC holds an intermediate arithmetic result in the teaching model; MDR holds a value or instruction being transferred between memory and the processor. A store address belongs in MAR, not automatically in ACC.

Answer 3.6

A hardware interrupt is a request from hardware for service. When an enabled request is accepted, the system saves the execution state it needs, transfers control to an interrupt service routine, then restores state and resumes. Masking, priorities and architecture affect when this occurs.

Answer 3.7

AND: 00→0, 01→0, 10→0, 11→1. OR: 00→0, 01→1, 10→1, 11→1. NOT: 0→1, 1→0. The usual AND symbol has a flat input side and rounded output side; OR has a curved input side; NOT is a triangle with an inversion circle. Named rectangular gate symbols are also valid if used consistently.

Chapter 4 — Networking

Learning outcome: Compare network architectures, clients and clouds; identify communication hardware and media; trace URLs, DNS and TCP/IP; explain application protocols, switching and network performance; distinguish data protection goals, encryption, signatures and TLS.

Prerequisites: Read the relevant earlier chapters; this chapter connects those ideas to networking.

Connect a room, a city, a world

What a network provides

A network connects devices so they can exchange data and share services, storage or hardware. A connected device is a node. Networks differ in scale, ownership, transmission media and management. Geographic size alone does not determine speed: a particular WAN link can be faster than a particular local wireless connection.

Names describe the scope

Network	Typical scope
PAN / WPAN	A person's nearby devices; a WPAN is wireless.
LAN / WLAN	A home, office or building; a WLAN uses wireless links.
MAN	A metropolitan area or several sites across a city.
WAN	Widely separated sites, regions or countries.

A practical analogy

A personal conversation resembles a PAN, a building intercom resembles a LAN, and an inter-city transport network resembles a WAN. Real networks can contain smaller networks inside them. The Internet is a global interconnection of networks, built from links and equipment operated by many organisations.

Apply and check

Try before reading feedback: Does lab LAN versus city WAN alone determine which has greater throughput?

Feedback: No. Link technology, capacity, congestion and endpoints determine actual performance; geography contributes propagation delay but is not a throughput specification.

Practice connection: Exercise 4.1a in exercises.md; explained answer in solutions.md.

Clients, servers & peers

Two ways to organise services

Client–server	Peer-to-peer
Clients request services from servers.	Peers can both request and provide resources.
Central management can simplify permissions and backup.	Resources can be distributed among participating peers.
Capacity and resilience require deliberate design.	Coordination and consistent security can be harder.

Servers are roles, not magic boxes

A file server provides access to shared files. One physical machine can host several server programs, and a service can run across many machines. A server failure need not stop an entire network if replication or failover is provided. A small P2P setup can be inexpensive, while large P2P systems can also scale when their protocols are designed for it.

A service request

Step / element	Meaning in the diagram
1	Client request
2	Network
3	Server response

A peer may act as a client for one exchange and as a server for another. Describe the role being performed.

Apply and check

Try before reading feedback: If the only campus dispatch server fails, what dependency has the architecture exposed?

Feedback: A central service can be a single point of failure unless replicated or supported by an appropriate local fallback.

Practice connection: Exercise 4.1b in exercises.md; explained answer in solutions.md.

Where does the work happen?

Thin and thick clients

Thin client	Thick client
Relies heavily on remote applications or services.	Performs substantial processing locally.
Can simplify central administration.	Can support more offline work.
Network and server availability affect its main tasks.	Needs suitable local hardware and maintenance.

Public and private clouds

A public cloud provides services using infrastructure operated by a provider for multiple customers. A private cloud is dedicated to one organisation and can be on-site or hosted elsewhere. Public cloud can reduce initial capital cost and offer flexible capacity; private cloud can provide greater direct control but requires resources to manage.

Compare actual responsibilities

Neither deployment is automatically secure. Consider access control, configuration, backups, data location, operational expertise and ongoing cost. Replication improves availability but is not a substitute for a recoverable backup. A thin client still performs some local processing, such as displaying an interface; “thin” does not mean it has no CPU.

Apply and check

Try before reading feedback: What breaks first if a thin client loses its server connection?

Feedback: Remote-dependent work may stop or degrade. A thick client may retain more local capability, though local storage and computation have costs.

Practice connection: Exercise 4.1d in exercises.md; explained answer in solutions.md.

The devices along a network path

Recognise the job

Device	Primary role
NIC	Connects a host to a network interface.
Switch / bridge	Forwards frames between LAN ports or segments.
Router	Forwards IP packets between networks.
Wireless access point	Connects wireless stations to a network.
Hub / repeater	Repeats signals; a hub sends them to multiple ports.
Gateway / modem	Translates between systems / modulates and demodulates signals.

Addresses serve different scopes

An Ethernet switch learns which MAC addresses are reachable through which ports. A router uses IP routing information to select a next hop. MAC addresses can be changed or randomised; they are not always permanently fixed identifiers. A modem's modulation depends on the medium, rather than being limited to old telephone lines.

A switch is selective, not omniscient

Known unicast frames can be sent to a selected port. Broadcasts and unknown destinations may be flooded within the relevant LAN. A router chooses a route using its routing table and policies; it does not always measure the least congested possible path for every packet.

Apply and check

Try before reading feedback: A packet leaves the campus subnet for another network. Is a switch alone the same as an IP router?

Feedback: No. A typical switch forwards local link-layer frames; a router forwards IP packets between networks. Some products combine roles.

Practice connection: Exercise 4.4a in exercises.md; explained answer in solutions.md.

Sharing a communication medium

Choose a link for its conditions

Twisted-pair copper reduces some electromagnetic interference by twisting paired conductors. Optical fibre carries light and supports high-capacity links. Wi-Fi uses shared radio channels; Bluetooth connects nearby devices using its own radio protocols. Range, speed and frequency-hopping behaviour depend on the version, equipment and environment, not one universal distance or channel count.

Collision detection and avoidance

CSMA/CD	CSMA/CA
Historically used on shared, half-duplex Ethernet.	Used by Wi-Fi to manage shared radio access.
Listen, transmit, detect collision, back off and retry.	Listen, wait/back off, transmit and use acknowledgements.
Modern switched full-duplex Ethernet has no collisions on the link.	RTS/CTS is an optional mechanism, not a compulsory step every time.

Use the conversation analogy

At a busy table, people listen before speaking and pause if they talk over each other. Random backoff reduces repeated clashes. Avoidance reduces the chance of a collision; it cannot guarantee one never happens. Interference, hidden stations and missing acknowledgements can still cause retries and delay.

Apply and check

Try before reading feedback: A modern full-duplex switched Ethernet link slows under load. Is collision detection necessarily the cause?

Feedback: No. That link does not normally use shared-medium collisions. Queueing, bottleneck capacity or endpoint limits can explain the delay.

Practice connection: Exercise 4.2 in exercises.md; explained answer in solutions.md.

Internet, web & addresses

The web is one Internet service

The Internet carries many kinds of communication. The World Wide Web uses linked resources accessed through web protocols. Email and other services also use the Internet. IPv4 addresses have 32 bits; IPv6 addresses have 128 bits. An address helps identify a network interface for communication, while routing determines how packets are forwarded.

Read a URL: https://example.org:443/course/notes?q=ai#part2

Part	Meaning
https	Scheme: how the resource is accessed.
example.org / :443	Host name / port number.
/course/notes	Path interpreted by the server.
?q=ai	Query supplied to the application.
#part2	Fragment identifying a location within the resource.

A URL is not just an IP address

The host name may be resolved through DNS. The path selects a resource on that host and need not correspond to a physical directory. Default ports can be omitted. A browser normally handles the fragment locally rather than sending it as part of an HTTP request.

Apply and check

Try before reading feedback: In https://campus.example/rover?id=7, which part is the host and which is the query?

Feedback: Host: campus.example. Query: id=7. HTTPS is the scheme; /rover is the path.

Practice connection: Exercise 4.3a in exercises.md; explained answer in solutions.md.

Find a host with DNS

Names are easier to remember

The Domain Name System maps names to records, including records that provide IP addresses. A browser and operating system may already have a cached answer. If not, a recursive resolver can obtain an answer using the DNS hierarchy. Cached answers remain usable for their permitted lifetime, reducing repeat lookups.

A simplified lookup

Step / element	Meaning in the diagram
1	Client
2	Recursive resolver
3	Root / TLD
4	Authoritative server

The resolver follows referrals, receives the requested record, caches it and returns the answer to the client.

The lookup is only one step

After resolving a host, a client still needs to establish suitable communication with the service. DNS does not deliver the web page itself, and it does not translate an entire URL, including its path, into an address. Failures in DNS, connectivity, transport, security negotiation or the application can each prevent a page from loading.

Apply and check

Try before reading feedback: DNS successfully returns an address but the web page still fails. Is that contradictory?

Feedback: No. Name resolution and application transfer are separate stages; routing, server availability or application errors can still fail.

Practice connection: Exercise 4.3b in exercises.md; explained answer in solutions.md.

Give each protocol layer a job

The four-layer TCP/IP model

Layer	Responsibility	Examples
Application	Application messages and services.	HTTP, DNS, SMTP
Transport	Communication between application endpoints.	TCP, UDP
Internet	IP addressing and routing between networks.	IPv4, IPv6, ICMP
Link	Local network delivery and access to the medium.	Ethernet, Wi-Fi

Encapsulation is packaging

Application data is carried inside transport data, which is carried inside IP packets, which are carried by link-layer frames. Each layer has its own fields and scope. A frame's MAC addresses are not the same thing as an IP packet's addresses; the frame can change at the next link while the packet continues its journey.

Reliable ordering is a service choice

TCP provides an ordered byte stream with reliability and flow/congestion control. UDP provides datagrams without those guarantees by itself. Applications may build additional mechanisms over UDP. Not every IP packet contains a sequence number that allows a router to rebuild a file: reconstruction belongs to the relevant endpoint protocol.

Apply and check

Try before reading feedback: At which TCP/IP layer does IP routing belong?

Feedback: The Internet layer. TCP belongs to transport; HTTP belongs to application; Wi-Fi is a link technology.

Practice connection: Exercise 4.4a in exercises.md; explained answer in solutions.md.

Web, files, email & sharing

Identify the application protocol

Protocol	Purpose
HTTP / HTTPS	Web requests and responses; HTTPS protects HTTP using TLS.
FTP	Traditional file transfer; plain FTP is not encrypted.
SMTP	Submitting and relaying email.
POP3	Retrieving email, with retention behaviour set by the client.
IMAP	Accessing and synchronising server-held mailboxes.
MIME	Structures message content and attachments; not a transport protocol.

A BitTorrent example

BitTorrent divides a file into pieces that peers exchange. A swarm is the participating group. A seed has the complete content and can upload pieces. A downloading peer may upload too; the word "leecher" does not necessarily mean a malicious user. Trackers can help peers discover one another, while distributed discovery is also possible.

Security is part of the choice

A protocol name alone may not guarantee confidentiality or authenticity. Use protected versions or secure alternatives where appropriate. File-transfer tools such as SFTP use a different protocol from FTP. A networked application must also authenticate users and apply access controls at the service.

Apply and check

Try before reading feedback: A student wants the same mailbox state on a laptop and phone. Which course protocol is the best fit?

Feedback: IMAP is designed to manage server-held mail across clients. Use an appropriately secured connection.

Practice connection: Exercise 4.4b in exercises.md; explained answer in solutions.md.

Reserve a path or share capacity?

Circuit and packet switching

Circuit switching	Packet switching
Reserves communication capacity for a session.	Shares links by sending discrete packets.
Can provide predictable allocated capacity.	Uses capacity efficiently across variable traffic.
Reservation can be idle during pauses.	Queues, loss and variable delay need management.

Packets move hop by hop

A sender divides or packages data according to its protocols. Routers forward IP packets using destination information. Packets may follow different routes or arrive out of order, although this is not required for every transfer. Endpoint protocols handle the reliability or ordering their applications need.

Apply the comparison carefully

A reserved circuit still has propagation and processing delay; it does not mean zero latency or perfect reliability. Modern Internet video calls commonly use packet-switched networks, buffering, congestion control and other techniques. The source exercise about a preferred circuit assumes a scenario valuing reserved capacity; it is not a universal description of today's video conferencing.

Apply and check

Try before reading feedback: Why can packet switching use a quiet user's spare capacity while a dedicated circuit cannot automatically do so?

Feedback: Packets statistically share links; a reserved circuit keeps allocated resources for its session even during idle periods.

Practice connection: Exercise 4.5c in exercises.md; explained answer in solutions.md.

Capacity is not responsiveness

Four problems to distinguish

Metric	What it tells you
Bandwidth / capacity	How much data a link can carry per second.
Throughput	The successful transfer rate actually achieved.
Latency	How long a transfer or response takes.
Jitter / loss	Variation in delay / missing packets.

Why a fast link can still feel slow

A wide road carries many cars, but a long journey still takes time. Similarly, high bandwidth does not remove propagation delay, queueing or server processing. Retransmitting lost data can add delay; interference and contention can reduce throughput. Measure the entire task, not just the advertised link rate.

Choose a response to the problem

Compress or summarise data when capacity is limited. Reduce unnecessary round trips when latency matters. Use timestamps to identify stale measurements. Buffering can smooth jitter for playback, but adds delay. Redundancy and reconnect logic can improve resilience, while local control keeps a time-critical action from depending on an unavailable remote link.

Apply and check

Try before reading feedback: A 100 Mb/s link has 200 ms command delay. Does the high capacity guarantee a responsive control loop?

Feedback: No. Capacity and latency are different. At 30 m/s, 0.2 s contributes 6 m of delay distance before response.

Practice connection: Exercise 6.5 in exercises.md; explained answer in solutions.md.

Protect data and its meaning

Three different goals

Goal	Question
Security	Who can access, change or disrupt the system?
Privacy	How is personal data collected, used and shared?
Integrity	Has the data remained correct and unaltered?

Validation versus verification

Validation checks whether an input meets rules: range, type, length or a valid calendar date. It cannot prove a plausible value is true. Verification checks that data agrees with a source or expected copy, such as double entry or a transfer checksum. Checksums detect some errors; a cryptographic authentication mechanism is needed to resist deliberate modification by an attacker.

Use layers of protection

Strong authentication, least-privilege access, updates, backups and network filtering address different risks. Malware can damage or steal data; spyware monitors activity; phishing tricks a user; pharming redirects a user to an unintended destination. A firewall is not a substitute for secure software, and antivirus cannot promise to detect every threat.

Apply and check

Try before reading feedback: A temperature is validly formatted as 99°C but the true reading is 29°C. Did format validation establish correctness?

Feedback: No. Validation establishes conformance to rules; verification against a reliable source or other checks is needed to find a plausible wrong value.

Practice connection: Exercise 4.6b in exercises.md; explained answer in solutions.md.

Keep messages confidential

From plaintext to ciphertext

Encryption uses an algorithm and a key to transform readable plaintext into ciphertext. Correctly implemented cryptography can protect confidentiality, but key management and authentication remain essential. Encryption alone does not prove who sent a message or ensure that a device receiving it is trustworthy.

Two approaches

Symmetric	Asymmetric
Uses a shared secret key.	Uses a related public/private key pair.
Efficient for bulk data encryption.	Supports mechanisms such as key establishment and signatures.
Both parties must obtain the secret securely.	The public key must be authenticated to its owner.

Combine their strengths

Secure protocols commonly use asymmetric mechanisms to authenticate or establish shared secrets, then symmetric authenticated encryption for application traffic. Keep private keys private. Do not invent an encryption scheme by substituting characters or treating a checksum as a secret; the design and implementation of cryptography require established protocols.

Apply and check

Try before reading feedback: Does a confidential encrypted packet automatically contain a safe command?

Feedback: No. Confidentiality hides data in transit. Correct authorisation, integrity/authenticity, freshness and command validation remain necessary.

Practice connection: Exercise 4.7a in exercises.md; explained answer in solutions.md.

Trust, signatures & certificates

A signature protects authenticity and integrity

A sender signs a message, usually through a hash-based construction, using a private signing key. A verifier checks the signature with the matching public key. A valid signature supports a claim about the key used and whether the signed content changed. It does not encrypt the message or prove that the message's claims are true.

A simplified verification path

Step / element	Meaning in the diagram
1	Message + signature
2	Trusted public key
3	Verify
4	Accept or reject

Signing is not generally "encryption in reverse"; the operations depend on the signature algorithm.

Certificates and TLS

A digital certificate binds an identity, such as a domain name, to a public key and is signed by an issuer. A client checks the trust chain, validity and expected name. TLS uses authentication and cryptography to protect communication, including HTTPS. SSL is obsolete and should not be deployed. A certificate can authenticate a domain without proving the site is honest in every other respect.

Apply and check

Try before reading feedback: An attacker replays an old message with its valid signature. What does the signature fail to establish?

Feedback: It does not establish freshness. Use appropriate sequence/session and timestamp or nonce checks alongside trusted signature verification.

Practice connection: Exercise 4.7c in exercises.md; explained answer in solutions.md.

Encryption modes & quantum ideas

Blocks and streams

A block cipher operates on fixed-size blocks; a mode specifies how it is used for longer messages. A stream cipher combines data with a generated keystream. Reusing a key/nonce combination incorrectly can undermine security. Authenticated encryption also detects unauthorised changes, rather than providing confidentiality alone.

Understand CBC as a historical example

Cipher block chaining combines a plaintext block with the previous ciphertext block before encryption, starting with an initialisation vector. Correct IV handling, padding and authentication matter. CBC is useful to understand, but a new application should use a vetted protocol and suitable authenticated-encryption construction rather than assembling its own cryptographic workflow.

What quantum key distribution does

QKD uses quantum signals and classical communication to establish key material. Its security analysis detects disturbance statistically under stated assumptions; it does not make every device or message unhackable. It requires specialised infrastructure and authenticated classical communication.
Post-quantum cryptography is a different approach: algorithms designed to resist quantum attacks on conventional computers.

Apply and check

Try before reading feedback: Why is a homemade encryption mode risky even if the block cipher is strong?

Feedback: Incorrect nonce handling, integrity protection or composition can break the system. Use vetted protocols and libraries with their documented requirements.

Practice connection: Exercise 4.7a in exercises.md; explained answer in solutions.md.

Chapter review

Complete ACT-04 using the supplied mission instructions. Then choose two related terms and explain a mechanism using both; identify one limitation of the teaching model and one condition that could change the result.

Reference support

• R3: RFC 1122 — Requirements for Internet Hosts: Communication Layers — Section 1.1.3 (Internet protocol suite) and the link, IP and transport sections. https://www.rfc-editor.org/rfc/rfc1122

• R4: RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3 — Section 1 (Introduction) and Section 8 (0-RTT and anti-replay). https://www.rfc-editor.org/rfc/rfc8446

Full resource introductions, optional tasks and access alternatives are in ../references.md.

Chapter 4 — Mission and optional resource

ACT-04 — The fast network sends a late warning

Chapter 4 — Networking

Pip's dashboard celebrates a high link rate. A stop warning arrives half a second late. "But the Wi-Fi is fast!" says the operator. The parcels keep moving.

Goal: Separate link capacity from response latency and calculate distance travelled during communication delay.

Preparation: Paper or calculator. Use a constant illustrative speed of 30 m/s and delays 500, 50, 5 and 200 milliseconds. This is not a campus-rover speed claim or a braking simulation.

Task:

1. Predict which delay causes the greatest extra travel before a response.

2. Convert each delay to seconds by dividing by 1,000. Calculate delay distance = speed × delay.

3. Create a two-column table of delay and distance. Explain why increasing Mb/s alone need not remove propagation, queueing or processing delay.

4. Optional command: python3 supporting/BlueImpulse_Demonstrations.py --topic network. Compare results and change the speed in your own calculation to 2 m/s.

Observe and record: Record units at each step and compare ratios: a ten-times smaller delay creates ten-times smaller delay distance at constant speed.

Reflect: What does the calculation omit? How could jitter or occasional long queueing delays matter even when average latency is low?

Feedback — check after trying: 500/50/5/200 ms correspond to 0.5/0.05/0.005/0.2 s and 15/1.5/0.15/6 m at 30 m/s. At 2 m/s they give 1/0.1/0.01/0.4 m. This excludes reaction after receipt, actuator lag and braking distance; do not call it total stopping distance.

Optional resource: R3 (full introduction and offline alternative follow).

R3 — RFC 1122 — Requirements for Internet Hosts: Communication Layers

IETF / R. Braden, editor, 1989.

https://www.rfc-editor.org/rfc/rfc1122

Relevant section: Section 1.1.3 (Internet protocol suite) and the link, IP and transport sections

Why use it: An authoritative foundation for the layered Internet model used in Chapter 4.

Learner task: Draw the four teaching layers and place HTTP, TCP, IP and Ethernet/Wi-Fi by role. Explain why a large link capacity does not set an end-to-end response time.

Limits: This is a technical historical standards document; current protocols have later updates. Use it for layer roles, not as a complete modern security checklist.

Offline alternative: The chapter and ACT-04 contain the layer explanations and all delay inputs/results offline.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

Chapter 4 — Networking

Key terms

Term	Meaning
LAN / WAN / MAN / PAN	Local / wide-area / metropolitan / personal network scopes.
Client / server / peer	Service requester / provider / participant that can perform both roles.
Thin / thick client	Relies more on remote resources / performs substantial local work.
Switch / router	Forwards local frames / forwards IP packets between networks.
NIC / access point	A network interface / a wireless network access device.
Protocol / stack	Communication rules / a layered organisation of those rules.
Frame / IP packet	A link-layer unit / a routed Internet-layer unit.
URL / DNS	A resource locator / a distributed name-resolution system.
TCP / UDP	Ordered reliable byte-stream transport / datagram transport.
Bandwidth / throughput	Link capacity / achieved successful transfer rate.
Latency / jitter	Delay / variation in delay.
Security / privacy / integrity	Protection / appropriate personal-data use / accuracy and resistance to alteration.
Validation / verification	Checking rules / checking agreement with a source or expected copy.
Plaintext / ciphertext	Readable original data / encrypted data.
Symmetric / asymmetric	Shared-secret mechanisms / public–private key mechanisms.
Signature / certificate	An authenticity-and-integrity check / an issuer-signed identity-to-key binding.
TLS / SSL	Current secure-transport protocol family / obsolete predecessor.
QKD	Key establishment using quantum communication with specific assumptions and infrastructure.

Chapter 4 — Networking

Exercises

Write reasoning, units and intermediate steps. All necessary concepts are explained in draft.md; the answer key is in solutions.md. The numbering preserves the source questions used throughout the guide.

Exercise 4.1a

Contrast LANs and WANs in geographic scale, ownership and typical transmission conditions. Avoid assuming that scale alone fixes speed.

Exercise 4.1b

Compare the client–server and peer-to-peer models in resource management and scalability.

Exercise 4.1c

Differentiate between a thick client and a thin client based on where processing and storage take place.

Exercise 4.1d

What are the main trade-offs in cost, security responsibilities and control between a public cloud and a private cloud?

Exercise 4.2

How do network systems handle simultaneous transmissions on a shared medium, using collision detection or collision avoidance?

Exercise 4.3a

Break down the parts of a URL, such as scheme, domain and path, and explain how it identifies a resource.

Exercise 4.3b

Describe the role of DNS when resolving the host name in a URL to an IP address.

Exercise 4.4a

Outline the four TCP/IP layers in order and give the core responsibility of each.

Exercise 4.4b

State the purpose of HTTP, FTP, POP3 and IMAP.

Exercise 4.5a

Explain what is meant by circuit switching.

Exercise 4.5b

Assume a video-conferencing scenario specifically requires reserved communication capacity. Explain why circuit switching could be preferred in that scenario. This is the source question’s assumed context, not a claim that modern Internet video calls generally use circuits.

Exercise 4.5c

List two benefits and one drawback of packet switching.

Exercise 4.6a

Differentiate between data security, data privacy and data integrity.

Exercise 4.6b

Contrast validation, such as range and format checks, with verification, such as double entry and transfer checks, when protecting data quality.

Exercise 4.6c

Name three mechanisms used to protect computer systems against threats.

Exercise 4.7a

Compare symmetric and asymmetric cryptography in key management, typical performance and the handling of plaintext and ciphertext.

Exercise 4.7b

Explain how a private signing key and matching public verification key support a digital signature that checks message origin and integrity. Distinguish signing from encryption.

Exercise 4.7c

What are SSL and TLS, and where is TLS used to secure network traffic today?

Chapter 4 — Networking

Explained answers

These are expected answers, not the only acceptable wording. Credit a correct alternative with clear assumptions and reasoning. Try the exercises before reading this section.

Answer 4.1a

A LAN connects a local site, often under one organisation's control. A WAN connects distant sites across wider geography and may depend on carriers. WAN paths often introduce longer propagation paths, but equipment, traffic and access links determine actual throughput and delay; scope alone does not guarantee a speed.

Answer 4.1b

Client–server centralises a service, making policy and backup management easier but creating capacity and availability responsibilities. Peers can both request and provide resources, spreading work but complicating discovery, trust and consistency. Both can scale with suitable designs; neither is automatically unlimited.

Answer 4.1c

A thick client performs substantial local processing and may keep local data. A thin client depends more on a remote server for computation and storage. Thin clients simplify some central management but depend more on network availability; actual systems can be hybrids.

Answer 4.1d

A public cloud uses provider-operated shared infrastructure, often lowering up-front costs but adding provider dependence and shared security responsibilities. A private cloud is dedicated to one organisation and offers more direct control, but the organisation must fund and operate or contract capacity and security. Neither is inherently secure merely because of its label.

Answer 4.2

Legacy shared, half-duplex Ethernet uses CSMA/CD to detect a collision and retry after backoff. Wi-Fi uses collision avoidance, including channel sensing, waiting and acknowledgements. Modern switched full-duplex Ethernet links normally avoid shared-medium collisions; congestion and queues can still occur.

Answer 4.3a

In https://example.org/guide/chapter1, https is the scheme, example.org is the host name and /guide/chapter1 is the path. A URL can also include a port, query and fragment. It identifies how and where to request a resource, not necessarily a fixed physical machine.

Answer 4.3b

DNS resolves names through a distributed hierarchy and caches. A resolver may return an address record for the requested host; the application can then use that address to contact a server. DNS does not itself download the web page, and cached records have limited lifetimes.

Answer 4.4a

Application: services and application messages. Transport: communication between application endpoints, such as TCP streams or UDP datagrams. Internet: IP addressing and routing across networks. Link/network-access: delivery over a local link, such as Ethernet or Wi-Fi. Layer names vary slightly between teaching models.

Answer 4.4b

HTTP requests and transfers web resources. FTP transfers files but its traditional form lacks confidentiality. POP3 retrieves email, often for local storage. IMAP manages and synchronises email kept on a server. Secure variants or TLS are required where credentials and content need protection.

Answer 4.5a

Circuit switching reserves communication resources along a path for a session. Capacity can remain allocated even when no useful data is being sent, giving a different sharing trade-off from statistical packet multiplexing.

Answer 4.5b

Under the stated reserved-capacity requirement, a circuit can provide predictable allocated capacity without competing for that same reservation during the call. It may waste idle capacity and needs setup. Modern Internet video calls normally use packet networks; the exercise is a hypothetical trade-off, not a claim about usual Internet practice.

Answer 4.5c

Benefits: bursty users can share link capacity efficiently, and packets can be forwarded along available paths. Drawback: queueing, loss or reordering can create variable delay and require handling. No route change or reliability is guaranteed merely by using packets.

Answer 4.6a

Security protects systems and data against unauthorised access or harm. Privacy concerns appropriate collection and use of personal information. Integrity concerns accuracy and resistance to unauthorised alteration. Encrypted personal data can still be collected without a legitimate purpose, so confidentiality alone does not settle privacy.

Answer 4.6b

Validation checks whether data satisfies rules such as format or range; a plausible wrong date can pass. Verification checks agreement with an original or expected copy, such as double entry or a transfer checksum. A checksum detects some changes but does not establish authenticity against an active attacker.

Answer 4.6c

Examples: least-privilege access controls, timely security updates, strong authentication, firewalls, encryption, protected backups and malware detection. Explain the threat addressed; no single mechanism stops every threat.

Answer 4.7a

Symmetric cryptography uses a shared secret and is efficient for bulk data, but securely distributing and protecting that secret matters. Asymmetric mechanisms use a public/private key pair for particular tasks such as key establishment or signatures. Both require correct algorithms and key handling; a public key is not itself secret.

Answer 4.7b

The sender signs with a private signing key; recipients verify with the matching public key. A valid signature supports origin and integrity relative to trust in that key, but does not hide the message. Encrypting for confidentiality is a distinct operation. A copied valid old signed message may still require replay protection.

Answer 4.7c

SSL is the obsolete predecessor to TLS. TLS is used today for protected transport in HTTPS and other application protocols, providing confidentiality and integrity with authentication according to configuration. TLS does not make an endpoint safe or prevent a legitimate user from misusing received data.

Chapter 5 — Applications

Learning outcome: Trace an integrated smart-system workflow; explain real-time deadlines and edge/cloud choices; connect policies to actuators; evaluate automation, human-AI interaction, ethical concerns and connected applications.

Prerequisites: Read the relevant earlier chapters; this chapter connects those ideas to applications.

A system that senses and acts

Combine the parts

An integrated smart system connects physical hardware, communication and software to carry out a useful task. Sensors collect information; processing interprets it and selects an output; actuators change the world. A network lets components exchange data or use remote services. Some systems use fixed rules, some use learned models, and some combine both.

Follow one complete workflow

Step / element	Meaning in the diagram
1	Sense
2	Communicate
3	Decide
4	Act

The changed environment supplies the next observation. The communication step may be local, remote or partly unnecessary for a self-contained controller.

Use the body analogy carefully

Sensors resemble senses, actuators resemble muscles and communication links resemble nerves. Software coordinates the response. This is an analogy for the roles, not a claim that the machine has a human mind. Explain the actual mechanism: what is measured, what message is sent, what computation runs and what physical output follows.

Apply and check

Try before reading feedback: The rover's sensor is correct but its motor cable is disconnected. Has a successful software decision completed the workflow?

Feedback: No. Physical actuation and feedback are required; logging a command is not proof that the environment changed.

Practice connection: Exercise 5.1 in exercises.md; explained answer in solutions.md.

Example: an irrigation system

Trace the flow of information

Stage	Illustrative design
Measure	Soil-moisture sensors report readings and timestamps.
Communicate	A low-power network sends readings to a local gateway.
Decide	A controller selects a watering action using targets and constraints.
Act	A driver energises a valve; later readings show the effect.

Add learning only when it has a role

A fixed threshold can open a valve when soil is dry. A learned policy might instead use moisture, recent rain and other state variables to trade off water use and crop needs. Reinforcement learning would require a defined reward and a process for updating the policy from experience. Merely connecting a sensor to Wi-Fi does not make the system learn.

Check assumptions before action

Sensors can drift; messages can arrive late; a valve can fail. Timestamp readings, detect implausible values and monitor the effect of each action. A real deployment needs domain-specific limits and testing. This example explains an architecture rather than prescribing watering durations for a particular crop.

Apply and check

Try before reading feedback: A moisture sensor sticks at "dry." What failure could a simple watering rule create?

Feedback: It could keep the valve open. Time/volume limits, plausibility checks and fault handling can contain the failure.

Practice connection: Exercise 5.3b in exercises.md; explained answer in solutions.md.

Real-time means meeting a deadline

Correctness includes timing

A real-time system must produce the required response within a specified deadline. "Real-time" does not simply mean very fast. Missing a hard deadline can make a system fail its requirements; missing a soft deadline reduces service quality. Batch work, such as a monthly report, usually has different timing needs.

Illustrative delay distance at 30 m/s

Delay before response	Distance travelled
500 ms = 0.500 s	30 × 0.500 = 15 metres
50 ms = 0.050 s	30 × 0.050 = 1.5 metres
5 ms = 0.005 s	30 × 0.005 = 0.15 metres

Interpret the calculation correctly

Distance during delay = speed × delay. This is only the distance travelled before a response begins, not total stopping distance. Sensing, processing, communication, actuation and physical braking all contribute to a real system. Moving a critical computation locally removes some communication delay, but it does not guarantee a particular safe outcome.

Apply and check

Try before reading feedback: A deadline is 100 ms. One response takes 120 ms, while the average is 40 ms. Was that response on time?

Feedback: No. Average speed does not excuse a missed specified deadline. The consequence depends on the task's real-time requirements.

Practice connection: Exercise 5.2a in exercises.md; explained answer in solutions.md.

Networking for a behaviour-based agent

Separate the fast loop from shared services

A useful architecture keeps time-critical sensing, policy evaluation and actuator control near the device. A network can carry telemetry, coordinate multiple devices or distribute reviewed model updates. Training may occur centrally while a fixed policy runs locally. A system can continue its local task during a link outage only if it was designed to do so.

Two connected responsibilities

Step / element	Meaning in the diagram
1	Device: observe / act
2	Network: data + updates
3	Server: analyse / train

Local action does not have to wait for every cloud exchange. This is an architectural choice, not a property of all AI systems.

Make messages interpretable

Include a device identity, timestamp, sequence number, units and model or policy version where needed. Reject stale or unauthorised commands. Delayed feedback must be associated with the action that produced it before a learning update is made. Network loss can reduce the evidence available for training; it should not silently become a reward of zero.

Apply and check

Try before reading feedback: Pip receives sequence 43 after 600 ms; the allowed age is 200 ms. Should it execute the command just because the ID is correct?

Feedback: No. Reject the stale command and use the defined safe local behaviour. Correct identity alone does not satisfy freshness.

Practice connection: Exercise 6.5 in exercises.md; explained answer in solutions.md.

Automation changes work and errors

A spectrum of responsibility

Mode	Who does what?
Manual	A human performs the task directly.
Assistance	The system supports selected actions or provides alerts.
Partial automation	The system handles parts of the task; the human supervises.
Higher autonomy	The system acts within a defined operating domain.

Benefits depend on the setting

Automation can improve repeatability, throughput and availability for suitable work. It can also shift errors from isolated human mistakes to systematic software, sensor or model failures affecting many operations. Do not assume that automation gives zero errors, unlimited uptime or a fixed percentage improvement.

Keep the operating domain explicit

A warehouse robot may work well on a mapped, maintained floor but fail on an unfamiliar surface. A human must understand what the system can handle, what it cannot handle and how a handover works. Formal driving-automation levels describe a specific domain; they should not be treated as a universal ladder for every smart appliance.

Apply and check

Try before reading feedback: What is the danger of automating an incorrect calibration rule across 500 devices?

Feedback: The same systematic error can be reproduced quickly at scale. Automation changes error patterns rather than eliminating them.

Practice connection: Exercise 5.2b in exercises.md; explained answer in solutions.md.

Feedback can change a policy

Describe the learning mechanism

An RL agent collects transitions: a state, an action, a reward and a next state. A learning algorithm uses those transitions to update values or policy parameters. A positive reward does not mechanically guarantee that an action becomes more likely in every algorithm; updates depend on the method and its estimates of future return.

Design an illustrative delivery task

Element	Possible definition
State	Location, remaining energy and relevant obstacle information.
Action	Choose a permitted move.
Reward	Score progress and successful completion; penalise undesirable outcomes.
Constraint	Disallow unsafe actions even if a reward would favour them.

A deployment may stop learning

A trained policy can be evaluated repeatedly on new observations without changing its parameters. Online learning requires a controlled update process and evidence that changes improve performance. Separate "the action changed because the observation changed" from "the policy changed because the agent learned".

Apply and check

Try before reading feedback: A reward arrives late. What association must be retained before using it to update learning?

Feedback: Associate it with the relevant episode, state/action or transition. Applying it to the latest unrelated action can corrupt learning.

Practice connection: Exercise 5.3a in exercises.md; explained answer in solutions.md.

Turn a decision into an action

A command needs an execution path

Step / element	Meaning in the diagram
1	Policy output
2	Controller
3	Power driver
4	Actuator

Feedback from position, current or other sensors can confirm whether the requested action happened.

Different physical outputs

Actuator	Example role
Electric motor / servo	Rotate a wheel or position a joint.
Solenoid valve	Open or close a fluid path.
Hydraulic actuator	Produce force using pressurised liquid.
Pneumatic actuator	Produce movement using compressed air.

Distinguish information from energy

A software value such as “target angle = 15 degrees” is information. Interface electronics convert it into a suitable control signal; a powered mechanism supplies the force. Some systems use analogue voltages, others PWM or digital commands. A DAC alone normally cannot supply motor power. Mechanical limits, response time and feedback all affect whether an intended action is achieved.

Apply and check

Try before reading feedback: Why should software read motor or position feedback after a move command?

Feedback: The motor may stall, saturate or fail. Feedback helps determine whether the intended physical outcome occurred.

Practice connection: Exercise 5.3b in exercises.md; explained answer in solutions.md.

Help people understand the system

Human-AI interaction is part of HCI

Human-computer interaction studies how people use computing systems. Human-AI interaction focuses on challenges raised by AI, including uncertainty, adaptation, delegation and trust. It is not a strict division between deterministic old interfaces and always-probabilistic new ones. The important question is what the person needs to understand and control.

Useful interface information

Show	Why it helps
Current mode and operating limits	The user knows what the system is doing.
Relevant evidence and uncertainty	The user can judge an output appropriately.
Planned action and reason	The next behaviour is less surprising.
Override, handover and recovery	The user can respond when the system cannot continue.

Avoid over-trust and under-trust

Unclear automation can cause misuse through excessive reliance, or disuse when useful assistance is rejected. A handover should communicate the state of the task and why control is changing. A confident-looking score is not enough: the interface should support an informed decision and an appropriate human response.

Apply and check

Try before reading feedback: A display says "AI approved" without uncertainty or override. Name a design improvement.

Feedback: State the relevant decision and its limits, show useful status, and provide an effective human override/recovery path appropriate to the risk.

Practice connection: Exercise 5.4b in exercises.md; explained answer in solutions.md.

Evaluate more than performance

Three important ethical concerns

Concern	Design question
Fairness	Who benefits, who is disadvantaged, and how was this measured?
Transparency	Can affected people understand the system's role and limitations?
Accountability	Who is responsible for deployment, oversight and correction?

Reward design can encode priorities

A reward that values only throughput can encourage neglect of other goals. Historical data can contain unequal representation or biased decisions. Evaluate outcomes across relevant groups and situations rather than assuming an algorithm is neutral because it uses numbers. The appropriate criteria depend on the task and the people affected.

Make responsibility practical

Document the intended use, data sources, limitations and monitoring process. Provide channels to question or correct consequential outcomes. Ethical and legal requirements vary by context and jurisdiction; no single checklist proves a system is responsible. For high-stakes uses, performance testing and meaningful human oversight are essential parts of the deployment decision.

Apply and check

Try before reading feedback: A routing model works worse for buildings with poor sensor coverage. What should the team investigate?

Feedback: Measure performance and impacts by relevant operating conditions, examine data coverage and accessibility, and adjust with accountable review.

Practice connection: Exercise 5.5a in exercises.md; explained answer in solutions.md.

Opportunities bring responsibilities

Illustrative connected applications

Application	Opportunity	Risk to consider
Traffic signals	Respond to changing demand.	Unsafe control or unequal service.
Water monitoring	Detect leaks and reduce waste.	Faulty sensors or unauthorised commands.
Street lighting	Adjust lighting to local need.	Privacy, coverage and maintenance.
Connected health monitoring	Support timely review of measurements.	Sensitive data and incorrect alerts.

Measure outcomes, not just activity

More sensor readings or more automated actions do not necessarily mean better service. Compare meaningful outcomes such as resource use, reliability, accessibility and user experience. Any claimed improvement needs evidence from the particular deployment; percentages from an unspecified example are not general guarantees.

Every connected device adds responsibilities

Connected cameras, locks, sensors and controllers need updates, access management and appropriate data protection. Minimise unnecessary personal data, limit retention and protect communication. Consider what happens if a device is compromised or the network is unavailable. Convenience should be assessed alongside maintainability, privacy and safe operation.

Apply and check

Try before reading feedback: Should a campus rover upload every camera frame merely because bandwidth is available?

Feedback: No. Determine a legitimate purpose, minimise data, consider privacy and retention, and use local processing where appropriate.

Practice connection: Exercise 5.5b in exercises.md; explained answer in solutions.md.

Chapter review

Complete ACT-05 using the supplied mission instructions. Then choose two related terms and explain a mechanism using both; identify one limitation of the teaching model and one condition that could change the result.

Reference support

• R1: Reinforcement Learning: An Introduction, second edition — Chapter 3 (finite Markov decision processes) and Section 6.5 (Q-learning). http://incompleteideas.net/book/the-book-2nd.html

• R4: RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3 — Section 1 (Introduction) and Section 8 (0-RTT and anti-replay). https://www.rfc-editor.org/rfc/rfc8446

• R5: Artificial Intelligence Risk Management Framework (AI RMF 1.0) — Framework functions Govern, Map, Measure and Manage; trustworthy AI characteristics.
https://www.nist.gov/itl/ai-risk-management-framework

Full resource introductions, optional tasks and access alternatives are in ../references.md.

Chapter 5 — Mission and optional resource

ACT-05 — Yesterday's command arrives today

Chapter 5 — Applications

Pip has stopped safely. A delayed "move" packet arrives and looks correctly formatted. Should it obey a message just because the bytes make sense?

Goal: Apply identity, unit, version, freshness and sequence checks while distinguishing secure transport from safe command handling.

Preparation: Receiver device ID robot-7; expected units m; policy_version 3; maximum age 200 ms; last accepted sequence 41. Assume the upstream authentication check is true for these examples. The toy Boolean is not a cryptographic implementation.

Task:

1. Inspect packets in order: A = sequence 42, age 50 ms; B = sequence 43, age 600 ms; C = sequence 42, age 60 ms. All other listed fields match.

2. For each packet, decide ACCEPT, REJECT STALE or REJECT DUPLICATE. Only update last accepted sequence when a packet is accepted.

3. Predict what would happen if a developer updated last sequence before checking age, and explain why reject paths should not corrupt accepted-state bookkeeping.

4. Optional command: python3 supporting/BlueImpulse_Demonstrations.py --topic networked-ai. Test the same order and compare receiver state.

Observe and record: Write the decision and last accepted sequence after each packet. Identify which check fails; do not invent a new packet timestamp.

Reflect: What must a real deployment add for clock assumptions, authentication, session resets and a safe local response when the network fails?

Feedback — check after trying: A is ACCEPT and advances last sequence to 42. B is STALE; last sequence stays 42. C is DUPLICATE; last sequence stays 42. Version, identity and unit checks add semantic protection. Real systems need authenticated messages, replay/session design, trustworthy timing and explicit fallback, not only this toy validation function.

Optional resource: R4 (full introduction and offline alternative follow).

R4 — RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3

IETF / Eric Rescorla, 2018.

https://www.rfc-editor.org/rfc/rfc8446

Relevant section: Section 1 (Introduction) and Section 8 (0-RTT and anti-replay)

Why use it: Distinguishes transport confidentiality/integrity from application-level decisions, including replay considerations.

Learner task: After ACT-05, explain why authenticated transport alone does not decide whether an old command is still appropriate to execute.

Limits: The Python receiver is a validation demonstration, not a TLS implementation. Do not implement cryptography from this worksheet.

Offline alternative: ACT-05 includes the packet fields, state transitions and rejection explanations; no network service or account is needed.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

Chapter 5 — Applications

Key terms

Term	Meaning
Smart system	Hardware, communication and software integrated for a useful task.
Real-time processing	Processing that must satisfy a specified timing deadline.
Automation	Technology performing some or all parts of a task with reduced direct human operation.
Edge / cloud computing	Processing near the source / processing on remote shared infrastructure.
Human-AI interaction	Interaction challenges involving AI, including uncertainty, delegation and trust.
Ethical AI	Design and use that consider fairness, accountability, transparency and impacts on people.
Operating domain	The conditions under which a system is intended to function.
Telemetry	Measurements and status information communicated for monitoring or analysis.

Chapter 5 — Applications

Exercises

Write reasoning, units and intermediate steps. All necessary concepts are explained in draft.md; the answer key is in solutions.md. The numbering preserves the source questions used throughout the guide.

Exercise 5.1

Define a smart system and explain how it integrates physical hardware, network connectivity and software into one workflow.

Exercise 5.2a

Define real-time processing and explain why communication delay and hardware performance matter for tasks such as collision avoidance.

Exercise 5.2b

What is automation, and how can moving from manual operation to automated systems change efficiency and error patterns?

Exercise 5.3a

Explain how an RL-based system uses feedback or reward signals from its environment to update its behavioural policy or value estimates.

Exercise 5.3b

Describe the role of actuators in translating a computed decision into a physical change.

Exercise 5.4a

Define human-AI interaction and explain the additional challenges it introduces within human-computer interaction.

Exercise 5.4b

Identify a usability or safety challenge that can occur when a smart system's autonomous behaviour lacks transparency.

Exercise 5.5a

Define ethical AI and discuss the challenges of maintaining transparency, fairness and accountability in smart systems.

Exercise 5.5b

Identify one opportunity and one security or privacy risk associated with connecting everyday physical devices to networks.

Chapter 5 — Applications

Explained answers

These are expected answers, not the only acceptable wording. Credit a correct alternative with clear assumptions and reasoning. Try the exercises before reading this section.

Answer 5.1

A smart system connects sensing, computation, communication and action to a goal. For irrigation, a moisture sensor supplies readings, software compares them with the policy, a network reports status and a valve changes water flow. It need not use machine learning simply because it is connected.

Answer 5.2a

Real-time processing means meeting a defined timing deadline, not merely being fast on average. Communication, queueing, computation and actuator delays all contribute. At 30 m/s, an extra 0.2 s delay means 6 m travelled before a response; this is delay distance, not total braking distance.

Answer 5.2b

Automation reduces direct manual operation of a task. It can improve consistency and throughput, but can also repeat a wrong rule at scale, fail on unusual inputs or change workers' monitoring responsibilities. Assess the actual process rather than assuming all errors disappear.

Answer 5.3a

An RL algorithm uses transitions and rewards to update policy parameters or value estimates, aiming at expected cumulative reward. A deployed agent may use a fixed learned policy without updating it on every action. Correctly associate feedback with the state and action that produced it.

Answer 5.3b

An actuator turns an electrical control signal into physical action: a motor moves a wheel, a valve changes flow, or a heater produces heat. The controller must respect limits and use feedback where needed; a software command is not proof the requested movement occurred.

Answer 5.4a

Human-AI interaction concerns how people work with AI-driven systems. Uncertainty, opaque decisions, changing performance and delegated actions add challenges to general interface design. Interfaces should communicate limits, support correction and provide meaningful human control.

Answer 5.4b

A user may over-trust an unexplained "safe" decision and ignore a hazard, or be unable to recover when an autonomous action is wrong. Display relevant status and uncertainty, explain the available action and offer an effective override or recovery process.

Answer 5.5a

Ethical AI considers consequences for people as well as performance: fairness across groups, explanations appropriate to users, privacy and accountable responsibility. Trade-offs must be evaluated in context, with monitoring and redress. A fairness score alone is not a complete ethical judgement.

Answer 5.5b

Opportunity: remote monitoring can find faults and reduce wasted energy. Risk: compromised devices can expose personal data or accept dangerous commands. Minimise collected data, authenticate access and maintain update and incident-response processes.

Chapter 6 — Strengths & Limitations

Learning outcome: Evaluate physical and network constraints; distinguish scalability, interoperability and resilience; compare predictive analytics with behavioural modelling; assess resource allocation, explainability, bias and responsible deployment.

Prerequisites: Read the relevant earlier chapters; this chapter connects those ideas to strengths & limitations.

Judge the whole system

Strengths and limitations interact

Part	Strength	Limitation
Hardware	Senses, computes and acts locally.	Power, heat, memory and physical reliability.
Networks	Share resources and coordinate devices.	Delay, loss, availability and security.
AI / control software	Selects actions and may learn from experience.	Model errors, limited evidence and unclear objectives.

There is no universally best architecture

Choose a design for its task, operating conditions and consequences of failure. A model with excellent offline accuracy may be unsuitable if it misses a deadline or requires too much memory. A highly connected system may be fragile if local operation depends on one remote service. A simple fixed-rule controller can be appropriate for a well-defined task.

Make the comparison explicit

State the intended outcome, identify constraints, compare options and define measures for success. Evaluate the combination of hardware, networking, algorithms and human operation. A component's specification is useful evidence, but it is not proof that the complete application meets its requirements.

Apply and check

Try before reading feedback: A model gains accuracy but misses the motor-control deadline. Is it automatically the better system?

Feedback: No. Evaluate the whole objective, including timing, safety, energy and accuracy in the operating domain.

Practice connection: Exercise 6.5 in exercises.md; explained answer in solutions.md.

Four physical budgets

Every device has limits

Step / element	Meaning in the diagram
1	Power
2	Heat
3	Memory
4	Processing

A design must fit all four budgets, not just the model's arithmetic requirement.

Understand the constraint

Budget	What can go wrong
Power	A battery drains too quickly for the required mission.
Heat	Thermal throttling reduces sustained performance.
Memory	Weights, inputs or intermediate results do not fit.
Processing	An inference or control computation misses its deadline.

Optimise with evidence

A smaller model, quantised representation, reduced input resolution or lower update rate can reduce cost, but may change accuracy or responsiveness. Measure the resulting behaviour. Dedicated accelerators can improve efficiency on supported workloads, but data conversion, memory movement and idle power remain relevant. Local computing removes a remote round trip; it does not mean zero total latency.

Apply and check

Try before reading feedback: Pip meets its deadline when cold but throttles after ten minutes. Which budget was overlooked?

Feedback: Thermal behaviour and sustained power. Test warm steady-state operation, not only short benchmark bursts.

Practice connection: Exercise 6.1 in exercises.md; explained answer in solutions.md.

Reliable decisions need reliable inputs

Sensors are imperfect

Noise creates random variation; drift changes a sensor's behaviour over time. Obstruction, calibration error or environmental conditions can produce misleading observations. Sensor fusion combines information from multiple sources, often after time alignment and filtering. Different sensors can have complementary strengths, but they can also share common failure causes.

A fusion pipeline

Step / element	Meaning in the diagram
1	Camera / range / motion
2	Align + check
3	Combine evidence
4	State estimate

Do not combine incompatible units or measurements from different times as if they described the same instant.

Define reliability under conditions

Reliability concerns performing an intended function for a stated period under stated conditions. MTBF is a statistical measure for repairable systems, not a promise that a particular unit cannot fail earlier. Monitor drift, error rates, temperature and data quality. Redundancy can help when failures are sufficiently independent and the system can recognise and manage disagreement.

Apply and check

Try before reading feedback: Two sensors disagree after months of use. Why should the controller consider drift?

Feedback: Calibration can change gradually. Compare reference measurements, inspect conditions and use a justified fault or recalibration process.

Practice connection: Exercise 6.1 in exercises.md; explained answer in solutions.md.

Grow without losing coordination

Three network qualities

Quality	Meaning
Scalability	Handle more devices or work at acceptable performance.
Interoperability	Exchange data and use it correctly across different systems.
Resilience	Maintain or recover an acceptable service during disruption.

Capacity is only one part of scaling

Adding hundreds of IoT devices can increase addressing, authentication, message traffic, storage and maintenance work. Agree on data formats, units, timestamps and interface versions as well as the physical link. Two products using Wi-Fi do not automatically interpret one another's messages correctly.

Plan for failures

Redundant links, retry policies, local buffering and controlled reconnection can improve resilience. A mesh can route around some failures when an alternative path exists and the protocols support it. It does not guarantee communication through every outage. Test overload and partial failure, not just the normal case with every device online.

Apply and check

Try before reading feedback: A receiver reads 250 centimetres as 250 metres. Is adding network capacity the main fix?

Feedback: No. It is a semantic interoperability failure. Specify and check units and schema/version information.

Practice connection: Exercise 6.2 in exercises.md; explained answer in solutions.md.

Put each computation where it fits

Three deployment patterns

Location	Benefit	Cost or limit
On-device edge	Low communication delay; local data.	Limited local resources and maintenance.
Nearby gateway / fog	Aggregates local devices.	Another service to size and manage.
Remote cloud	Shared compute and flexible capacity.	Network dependency, latency and transfer cost.

An illustrative warehouse design

Each robot handles immediate motion control locally. A shared service allocates tasks and coordinates inventory. Robots send selected telemetry and receive authorised updates. If the coordination service is unavailable, the design defines whether a robot pauses, completes a limited task or enters another safe state; continued operation must not be assumed.

Optimise across the whole path

Compare local compute energy with transmission energy, available bandwidth, deadlines, privacy and model capacity. Summarising data locally can reduce transfer volume. Centralised analysis can reveal patterns across devices. The useful decision is which part of the workload belongs where, not whether edge or cloud is always superior.

Apply and check

Try before reading feedback: Choose a location for emergency stopping and a location for weekly fleet analytics, and justify each.

Feedback: Local control avoids remote round-trip dependence for stopping; cloud or central processing can aggregate fleet history for slower analytics. Both need security and resource planning.

Practice connection: Exercise 6.5 in exercises.md; explained answer in solutions.md.

Predict an outcome or model behaviour?

Two useful questions

Predictive analytics	Behavioural modelling
What is likely to happen?	How does an agent or system behave?
Forecasts, scores or probability estimates.	Rules, transitions, trajectories or action patterns.
Example: estimate a component's failure risk.	Example: model a robot's response to an obstacle.

The approaches can overlap

A behavioural model can be used to make predictions, and predictive features can inform an action policy. They are not mutually exclusive categories defined by one kind of data. Choose the model and evaluation method based on the question being asked and the evidence available.

Adaptation is not guaranteed generalisation

Adaptive intelligence describes changing behaviour appropriately using new information or experience. Autonomy describes operating with limited direct human control. An autonomous device may use a fixed policy; an adaptive model may still need supervision. Performance in familiar conditions does not prove reliability in a new environment.

Apply and check

Try before reading feedback: A model predicts a busy corridor at noon. Has it proved why students choose that corridor?

Feedback: No. A predictive association can be useful without a causal explanation. Behavioural modelling may represent decisions, but also needs justified assumptions.

Practice connection: Exercise 6.4 in exercises.md; explained answer in solutions.md.

Understand why a policy acted

Explainability is a practical capability

Explainability concerns making relevant aspects of a model's behaviour understandable to people. A small rule set or Q table may be inspectable. A deep RL policy can be harder to interpret because many parameters contribute to each output. Complexity does not mean analysis is impossible, but a simple story about one decision may be incomplete or misleading.

Look beyond one successful outcome

Inspect representative cases, counterexamples, changed conditions and failure logs. Separate a faithful explanation of the computation from a plausible narrative generated after the fact. An explanation does not establish that the action was correct; validation requires evidence about outcomes and requirements.

Reward errors and bias

An agent rewarded only for speed may neglect an important constraint. Training environments can under-represent some conditions or groups. Distribution shift can reduce performance when deployment differs from training. Track both overall performance and relevant subsets. Correcting a reward or dataset requires re-evaluation of the resulting policy.

Apply and check

Try before reading feedback: A heat map highlights an obstacle. Does it prove the model used exactly the reasoning a human expects?

Feedback: No. It is an explanatory aid with limits, not conclusive access to the model's causal decision process.

Practice connection: Exercise 6.6 in exercises.md; explained answer in solutions.md.

Balance benefit, risk & oversight

Questions for a deployment review

Area	Evidence to seek
Security	Authenticated updates, restricted access and recovery plans.
Privacy	Necessary data collection, appropriate retention and access.
Fairness	Evaluation across relevant people and conditions.
Accountability	Named responsibilities, review and correction routes.
Reliability	Measured deadlines, failure handling and monitoring.

Physical effects raise the stakes

A connected controller can change the physical world, so a corrupted measurement or unauthorised command may have consequences beyond a damaged file. Protect sensor inputs, model updates and actuator commands according to the task. Test how the system responds to missing, stale and contradictory data.

Finish with a justified judgement

State the benefit you expect, the evidence supporting it, the conditions under which it applies and the remaining limitations. Decide what a human must monitor or approve and how the system should respond when its assumptions fail. The strongest design is the one suited to its purpose, with limitations that are understood and managed.

Apply and check

Try before reading feedback: Pip succeeds on one sunny test run. Name two further checks before claiming robust deployment.

Feedback: Test changed lighting/weather or sensor faults, and evaluate adverse traffic/network conditions with monitored fallback. A single success does not establish reliability across the operating domain.

Practice connection: Exercise 6.7 in exercises.md; explained answer in solutions.md.

Chapter review

Complete ACT-06 using the supplied mission instructions. Then choose two related terms and explain a mechanism using both; identify one limitation of the teaching model and one condition that could change the result.

Reference support

• R5: Artificial Intelligence Risk Management Framework (AI RMF 1.0) — Framework functions Govern, Map, Measure and Manage; trustworthy AI characteristics.
https://www.nist.gov/itl/ai-risk-management-framework

Full resource introductions, optional tasks and access alternatives are in ../references.md.

Chapter 6 — Mission and optional resource

ACT-06 — Choose a system, not a leaderboard

Chapter 6 — Strengths & Limitations

A larger model wins an accuracy contest but overheats Pip's computer. A cloud model is accurate too, until the network drops. The team must choose a deployment, not a single impressive number.

Goal: Evaluate edge/cloud placement against explicit latency, memory, power, privacy and resilience constraints.

Preparation: Use this fictional design table: local small model = 30 ms compute, 180 MB working memory, 4 W, 92% measured task accuracy; local large model = 90 ms, 700 MB, 9 W, 96%; remote model = 20 ms server compute plus 100–300 ms round trip, 98%, uploads images. Device budgets: 512 MB, 6 W and 80 ms deadline. Treat accuracies as supplied hypothetical measurements, not research results.

Task:

1. Eliminate designs that violate hard local memory, power or timing budgets. For the remote model add the network and server times.

2. Choose where emergency control and overnight fleet analytics should run. State how the system behaves during lost connectivity.

3. A software update improves the small model to 94% but needs 70 ms and 5 W. Decide what to test before accepting it; do not assume that one benchmark settles safety.

4. Draw a two-column decision record: constraint/evidence, then choice/trade-off. Propose one privacy safeguard and one monitoring measure.

Observe and record: Separate hard failures from preferences and record the worst stated time rather than only the mean. Identify unknowns such as warm-device timing and test-data coverage.

Reflect: Why is 98% accuracy not sufficient to choose the remote model for the 80 ms task? Which human decision remains after the calculations?

Feedback — check after trying: The small local model fits the stated budgets. The large local model violates memory, power and latency. The remote option takes 120–320 ms and misses the deadline; it may suit slower analytics with privacy controls. The updated small model fits the listed limits but leaves only 10 ms timing margin, requiring end-to-end/warm-device testing, relevant accuracy evaluation and a fallback. Valid alternatives earn credit if their assumptions and trade-offs are explicit.

Optional resource: R5 (full introduction and offline alternative follow).

R5 — Artificial Intelligence Risk Management Framework (AI RMF 1.0)

NIST, NIST AI 100-1, January 2023.

https://www.nist.gov/itl/ai-risk-management-framework

Relevant section: Framework functions Govern, Map, Measure and Manage; trustworthy AI characteristics

Why use it: Connects measured performance with context, impacts, monitoring and responsibility.

Learner task: For ACT-06, assign one concrete action to each framework function: identify an owner, describe the operating domain, measure a relevant failure and plan a response.

Limits: A voluntary risk-management framework is not certification that a product is safe, fair or legally compliant. Context and evidence still matter.

Offline alternative: ACT-06 supplies the design table and expected trade-off reasoning. A suitable four-action response is: Govern — name an accountable operator; Map — define the campus operating domain; Measure — test missed deadlines and uneven accuracy; Manage — maintain fallback and an incident response process.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

Chapter 6 — Strengths & Limitations

Key terms

Term	Meaning
Sensor integration	Connecting and coordinating measurements for a computing system.
Hardware constraints	Limits such as power, thermal capacity, memory and compute time.
Reliability	Performance of the intended function over time under stated conditions.
Sensor drift	A gradual change in measurement behaviour or calibration.
Scalability	Ability to handle growth at acceptable performance.
Interoperability	Ability to exchange information and use it correctly across systems.
Resilience	Ability to maintain or recover acceptable service through disruption.
Resource optimisation	Balancing computation, communication, energy and other costs for a goal.
Adaptive intelligence	Changing behaviour appropriately using new information or experience.
Autonomy	Operating with limited direct human intervention.
Predictive analytics	Using evidence to forecast outcomes.
Behavioural modelling	Representing or analysing behaviour and interactions.
Explainability	Making relevant aspects of model behaviour understandable.
Bias / distribution shift	Systematic distortion / a difference between training and deployment conditions.

Chapter 6 — Strengths & Limitations

Exercises

Write reasoning, units and intermediate steps. All necessary concepts are explained in draft.md; the answer key is in solutions.md. The numbering preserves the source questions used throughout the guide.

Exercise 6.1

Explain how thermal limits, power consumption, memory capacity and processing speed constrain complex AI models on small embedded devices.

Exercise 6.2

Define scalability and interoperability. Explain why both matter when adding hundreds of IoT devices from different manufacturers.

Exercise 6.3

Define adaptive intelligence and explain how it can help an agent respond to changing conditions, without assuming that adaptation guarantees success.

Exercise 6.4

Compare predictive analytics and behavioural modelling in how they use evidence to inform future decisions or actions.

Exercise 6.5

Explain resource optimisation when balancing local hardware processing against network transmission costs.

Exercise 6.6

Define explainability and explain why some deep reinforcement-learning models are difficult to interpret.

Exercise 6.7

Identify two security concerns and two ethical implications when a network of smart devices makes high-stakes decisions with limited human oversight.

Chapter 6 — Strengths & Limitations

Explained answers

These are expected answers, not the only acceptable wording. Credit a correct alternative with clear assumptions and reasoning. Try the exercises before reading this section.

Answer 6.1

An embedded device has limited memory for weights and working data, compute time for deadlines, power from its supply and thermal capacity to remove heat. A model that fits storage can still exceed RAM or latency budgets. Consider quantisation or a smaller model, then measure accuracy and worst-case timing on the target device.

Answer 6.2

Scalability means accommodating growth while meeting acceptable service targets. Interoperability means exchanging and correctly using information between systems. A network that can carry many packets still fails interoperability if one sensor sends centimetres and the receiver assumes metres. Test load, schemas, units, versions and identities.

Answer 6.3

Adaptive intelligence changes behaviour appropriately using new evidence or experience. It can respond to changed conditions, but noisy feedback, distribution shifts or unsafe exploration can make adaptation harmful. Test changes against objectives and retain monitoring and recovery controls.

Answer 6.4

Predictive analytics estimates future or unknown outcomes from evidence, such as a likely equipment failure. Behavioural modelling represents actions and interactions, such as how drivers respond to signs. A model can be useful without proving causes; both require relevant data and evaluation on new cases.

Answer 6.5

Resource optimisation selects where and how to compute while balancing latency, accuracy, energy, bandwidth, privacy and cost. Local emergency control can avoid a remote round trip; cloud processing can support large-scale training or reporting. Edge processing still has hardware limits and requires maintenance.

Answer 6.6

Explainability makes relevant aspects of model behaviour understandable to a particular audience. Deep RL can be difficult to explain because many learned parameters and sequential consequences interact. An approximate post-hoc explanation may be useful, but is not a guarantee of the model's true causal reasoning.

Answer 6.7

Security concerns include unauthorised commands and replay or manipulation of sensor messages. Ethical concerns include unfair impacts, privacy loss and unclear responsibility for harm. Combine authentication, freshness checks, monitoring, human oversight and explicit accountability. Valid alternative risks and controls should earn credit when well justified.

Blue Impulse — References and resource introductions

The six team-supplied chapter PDFs defined the scope. Their notes credit Cambridge International AS & A Level Computer Science but do not identify a full edition or author bibliography; this remains a provenance limitation and no edition has been invented. The new activities, calculations and checkpoints are original AI-assisted teaching examples. The guide's original artwork is user supplied and retains its own third-party rights.

The resources below were inspected as optional extensions and reference support. No external link replaces an explanation, input, question or answer in this snapshot. No student-created website or external platform is required.

R1 — Reinforcement Learning: An Introduction, second edition

Richard S. Sutton and Andrew G. Barto, MIT Press, 2018.

http://incompleteideas.net/book/the-book-2nd.html

Relevant section: Chapter 3 (finite Markov decision processes) and Section 6.5 (Q-learning)

Why use it: Primary textbook support for the agent/environment distinction, expected return and the Q-learning update.

Learner task: After ACT-02, explain in your own words what the bootstrapped target includes, then compare it with the terminal-state case.

Limits: The book is mathematically deeper than this beginner guide. Start with the supplied worked calculation; do not treat the browser page as required activity content.

Offline alternative: The formula, both complete calculations and interpretation are included in ACT-02 and the chapter, so the task works offline.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

R2 — Nand to Tetris — Project 5: Computer Architecture

Noam Nisan and Shimon Schocken / Nand to Tetris.

https://www.nand2tetris.org/project05

Relevant section: Project overview, proposed computer architecture and hardware simulator guidance

Why use it: A concrete educational computer architecture connects logic, CPU state and memory.

Learner task: After ACT-01, compare the purpose of an instruction address and a data-memory address with the project's computer model. Name one part the models share and one detail that differs.

Limits: The Hack computer is a different ISA; its instructions and registers must not be substituted directly into this guide's toy program. Software installation is optional and not required for the paper task.

Offline alternative: ACT-01 includes its full program, trace and expected output. ACT-03 provides a complete paper pipeline grid specification.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

R3 — RFC 1122 — Requirements for Internet Hosts: Communication Layers

IETF / R. Braden, editor, 1989.

https://www.rfc-editor.org/rfc/rfc1122

Relevant section: Section 1.1.3 (Internet protocol suite) and the link, IP and transport sections

Why use it: An authoritative foundation for the layered Internet model used in Chapter 4.

Learner task: Draw the four teaching layers and place HTTP, TCP, IP and Ethernet/Wi-Fi by role. Explain why a large link capacity does not set an end-to-end response time.

Limits: This is a technical historical standards document; current protocols have later updates. Use it for layer roles, not as a complete modern security checklist.

Offline alternative: The chapter and ACT-04 contain the layer explanations and all delay inputs/results offline.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

R4 — RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3

IETF / Eric Rescorla, 2018.

https://www.rfc-editor.org/rfc/rfc8446

Relevant section: Section 1 (Introduction) and Section 8 (0-RTT and anti-replay)

Why use it: Distinguishes transport confidentiality/integrity from application-level decisions, including replay considerations.

Learner task: After ACT-05, explain why authenticated transport alone does not decide whether an old command is still appropriate to execute.

Limits: The Python receiver is a validation demonstration, not a TLS implementation. Do not implement cryptography from this worksheet.

Offline alternative: ACT-05 includes the packet fields, state transitions and rejection explanations; no network service or account is needed.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

R5 — Artificial Intelligence Risk Management Framework (AI RMF 1.0)

NIST, NIST AI 100-1, January 2023.

https://www.nist.gov/itl/ai-risk-management-framework

Relevant section: Framework functions Govern, Map, Measure and Manage; trustworthy AI characteristics

Why use it: Connects measured performance with context, impacts, monitoring and responsibility.

Learner task: For ACT-06, assign one concrete action to each framework function: identify an owner, describe the operating domain, measure a relevant failure and plan a response.

Limits: A voluntary risk-management framework is not certification that a product is safe, fair or legally compliant. Context and evidence still matter.

Offline alternative: ACT-06 supplies the design table and expected trade-off reasoning. A suitable four-action response is: Govern — name an accountable operator; Map — define the campus operating domain; Measure — test missed deadlines and uneven accuracy; Manage — maintain fallback and an incident response process.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

R6 — The Unicode Standard — technical introduction

The Unicode Consortium.

https://www.unicode.org/standard/principles.html

Relevant section: Unicode design principles and character encoding model

Why use it: Supports the difference between a code point and a variable-length encoding such as UTF-8.

Learner task: Explain why one character is not necessarily one byte, and why BCD digits and text digits need an agreed representation.

Limits: Full text processing includes combining characters and grapheme clusters beyond this chapter. A displayed symbol is not always one code point.

Offline alternative: Chapter 1 defines character sets, code points and encodings and includes the BCD/ordinary-binary comparison.

Access: Optional public web resource; no account required to read the cited page. Signed-out HTTP access was checked on 7 November 2026. Availability on every campus network is not guaranteed. All essential task instructions and feedback are supplied locally.

Responsible AI-use record

Neogpt/Codex assisted on 7 November 2026 with reconciling the guide into Markdown, developing activities and answers, drafting the Test Bank, generating PDFs/DOCX and verifying package structure. The user asked to follow rules/Stage2_All_Instructions.md. Calculations and supplied code outputs were checked locally; source/resource access and document checks are recorded in the submission evidence. These checks are not independent human peer review. Significant interaction excerpts belong in the project and personal logs.

Blue Impulse — Stage 2 Test Bank V1.0

Module: Hardware, Networking and Behaviorist AI
Version/date: V1.0 — 7 November 2026
Coverage: Six chapters; 24 True/False, 24 single-answer Multiple Choice and 6 open-ended tasks.

Instructions

Attempt the questions before reading the answer key. For True/False, correct any false statement. For multiple choice, select exactly one answer. Open responses require developed reasoning: normally at least 120 words (150 for OE06), or an equivalent labelled diagram, decision matrix, calculations or worked trace with explanations. Every question includes its own required numerical data; no online platform is needed. Suggested self-assessment: 1 mark for each T/F decision plus 1 for reasoning; 1 mark for each MC choice plus 1 for reasoning; 10 marks for each open response. Total 156 marks. This is a practice allocation, not a teacher-issued grading scheme.

Questions — True/False

TF01 — Chapter 1: The Fundamentals / Represent negative values

In eight-bit two's complement, the representable range is −128 to +127.

TF02 — Chapter 1: The Fundamentals / Numbers, characters & codes

Unicode requires every character to occupy exactly one byte.

TF03 — Chapter 1: The Fundamentals / The stored-program computer

A 3 GHz clock guarantees exactly three billion completed instructions every second.

TF04 — Chapter 1: The Fundamentals / Trace a tiny program

In the guide's toy program, STORE 200 copies ACC into memory address 200 without making 200 the result.

TF05 — Chapter 2: Behaviorism & Behaviorist AI / React first: behaviour-based robots

A fixed thermostat rule that never changes from experience is, by itself, an example of reinforcement learning.

TF06 — Chapter 2: Behaviorism & Behaviorist AI / Rewards, policies & values

An RL policy can prefer an action with a small immediate cost when it offers greater expected future return.

TF07 — Chapter 2: Behaviorism & Behaviorist AI / Update a Q value

For a terminal next state in the guide's Q-learning update, the future-value term is zero.

TF08 — Chapter 2: Behaviorism & Behaviorist AI / Training is different from using a policy

Every action chosen by a deployed learned policy necessarily changes its trained parameters.

TF09 — Chapter 3: Hardware / Memory: capacity versus speed

SRAM and DRAM normally retain their data after all power is removed.

TF10 — Chapter 3: Hardware / Overlap work with a pipeline

An ideal four-stage pipeline can finish three independent instructions in six cycles while each instruction still traverses four stages.

TF11 — Chapter 3: Hardware / The hardware behind an AI model

A GPU is guaranteed to beat a CPU on every program because it has many parallel execution units.

TF12 — Chapter 3: Hardware / Sense, decide, act, sense again

A fan controller can use different switch-on and switch-off thresholds to reduce rapid toggling near one temperature.

TF13 — Chapter 4: Networking / Capacity is not responsiveness

High network bandwidth alone guarantees low command latency.

TF14 — Chapter 4: Networking / Sharing a communication medium

Modern switched full-duplex Ethernet links normally avoid the shared-medium collisions that CSMA/CD handles.

TF15 — Chapter 4: Networking / Trust, signatures & certificates

A digital signature by itself keeps a message confidential.

TF16 — Chapter 4: Networking / Find a host with DNS

Successful DNS resolution does not guarantee that the web server will return a requested page.

TF17 — Chapter 5: Applications / Real-time means meeting a deadline

A system meets every 100 ms deadline if its average response time is 40 ms, even when one response takes 120 ms.

TF18 — Chapter 5: Applications / Networking for a behaviour-based agent

A correctly authenticated message can still be too old to execute safely.

TF19 — Chapter 5: Applications / Feedback can change a policy

Delayed reward should always be assigned to the most recently received action, regardless of its origin.

TF20 — Chapter 5: Applications / Turn a decision into an action

A logged motor command does not prove that the requested physical movement occurred.

TF21 — Chapter 6: Strengths & Limitations / Four physical budgets

A model that fits in storage automatically fits the device’s working-memory, timing, power and thermal budgets.

TF22 — Chapter 6: Strengths & Limitations / Grow without losing coordination

Agreeing on units and schema versions supports interoperability between different devices.

TF23 — Chapter 6: Strengths & Limitations / Put each computation where it fits

Moving every computation to the cloud always improves resilience during network loss.

TF24 — Chapter 6: Strengths & Limitations / Understand why a policy acted

A useful post-hoc explanation is not necessarily a faithful account of the exact causal process that produced a model's decision.

Questions — Multiple Choice

MC01 — Chapter 1: The Fundamentals / From bits to bytes

How many bytes are in 2 KiB?

A. 16,384

B. 2,000

C. 16,000

D. 2,048

MC02 — Chapter 1: The Fundamentals / Build a digital image

A 500 × 200 bitmap uses 16 colours and no padding. What is its raw pixel-data size, excluding headers and compression?

A. 100,000 bytes

B. 12,500 bytes

C. 25,000 bytes

D. 50,000 bytes

MC03 — Chapter 1: The Fundamentals / Fetch / decode / execute

Which register identifies the memory location being accessed in the teaching model?

A. CIR

B. ACC

C. MAR

D. MDR

MC04 — Chapter 1: The Fundamentals / Trace a tiny program

Starting with ACC = 0, run LOAD #3; ADD #9; STORE 200; HALT. What is memory[200]?

A. 12

B. 9

C. 200

D. 3

MC05 — Chapter 2: Behaviorism & Behaviorist AI / The reinforcement-learning loop

Which description best defines a policy?

A. The numerical score after a transition

B. A strategy for choosing actions from a state or observation

C. A guarantee that every action succeeds

D. The physical environment around the agent

MC06 — Chapter 2: Behaviorism & Behaviorist AI / Update a Q value

Old Q = 2, alpha = 0.5, reward = 1, gamma = 0.9, best next Q = 4. What is the updated non-terminal Q value?

A. 2.6

B. 1.3

C. 3.3

D. 4.6

MC07 — Chapter 2: Behaviorism & Behaviorist AI / Explore or exploit?

Pip always chooses the route that looked best in its first two trials. What is the most direct learning concern?

A. It may never gather evidence about a better untried route

B. It must receive a negative reward every time

C. It is guaranteed to explore every route equally

D. It must have a perfect transition model

MC08 — Chapter 2: Behaviorism & Behaviorist AI / Actions can have uncertain outcomes

An action gives reward +10 with probability 0.8 and −5 with probability 0.2. What is expected immediate reward?

A. 8

B. 5

C. 7

D. 10

MC09 — Chapter 3: Hardware / Memory: capacity versus speed

Which description correctly distinguishes DRAM from SRAM?

A. DRAM charge needs refresh; powered SRAM bistable cells do not need periodic refresh

B. SRAM is a magnetic disk technology

C. SRAM can only store instructions and not data

D. DRAM is always non-volatile

MC10 — Chapter 3: Hardware / Overlap work with a pipeline

An ideal four-stage pipeline processes five independent instructions with no stalls. How many cycles elapse from the first fetch through the last write-back?

A. 8

B. 4

C. 20

D. 5

MC11 — Chapter 3: Hardware / Reason with logic gates

Movement is allowed when battery_ok AND NOT obstacle. What is the output for battery_ok = 1, obstacle = 1?

A. It must alternate

B. 1

C. 0

D. The expression is undefined

MC12 — Chapter 3: Hardware / The hardware behind an AI model

Which task most naturally exposes SIMD-style parallelism?

A. Following a fully dependent branch chain

B. Executing a single unpredictable pointer dependency

C. Waiting for one keyboard key

D. Applying the same arithmetic operation to many image pixels

MC13 — Chapter 4: Networking / Capacity is not responsiveness

At 30 m/s, how far is travelled during 50 ms of communication delay alone?

A. 0.15 m

B. 1.5 m

C. 150 m

D. 15 m

MC14 — Chapter 4: Networking / Give each protocol layer a job

In the four-layer TCP/IP teaching model, where does IP routing belong?

A. Link/network-access

B. Transport

C. Application

D. Internet

MC15 — Chapter 4: Networking / Protect data and its meaning

Which check primarily establishes that data obeys an allowed format or range?

A. A device driver

B. Validation

C. Lossy compression

D. Encryption

MC16 — Chapter 4: Networking / Trust, signatures & certificates

Which combination best describes how a digital signature is created and checked?

A. Encrypt every message with a password shown inside it

B. Sign with a private signing key; verify with the matching public key

C. Sign by hashing alone; no key is involved

D. Sign with a public key; hide with the same public key

MC17 — Chapter 5: Applications / Real-time means meeting a deadline

A local task has a maximum permitted response of 80 ms. Which listed response meets that deadline?

A. 100 ms

B. 60 ms

C. An average of 40 ms with this response at 90 ms

D. 120 ms

MC18 — Chapter 5: Applications / Networking for a behaviour-based agent

Receiver last accepted sequence is 42; allowed age is 200 ms. A matching, authenticated packet has sequence 43 and age 600 ms. What should the toy receiver do?

A. Accept because the sequence is new

B. Reject stale and keep the accepted sequence at 42

C. Accept because authentication overrides age

D. Reset the accepted sequence to 0

MC19 — Chapter 5: Applications / Turn a decision into an action

Which component directly changes water flow in an automated irrigation system?

A. Valve actuator

B. Moisture sensor

C. DNS resolver

D. Reward value

MC20 — Chapter 5: Applications / Feedback can change a policy

What is needed before using a delayed reward in a learning update?

A. Associate it with the relevant transition or episode

B. Discard all earlier records forever

C. Assume it belongs to whichever action arrived last

D. Replace every negative reward with zero

MC21 — Chapter 6: Strengths & Limitations / Four physical budgets

Pip has 512 MB working memory and a 6 W budget. Which design meets both stated limits?

A. 700 MB and 4 W

B. 700 MB and 9 W

C. 180 MB and 4 W

D. 400 MB and 9 W

MC22 — Chapter 6: Strengths & Limitations / Grow without losing coordination

A sensor sends 250 centimetres, but a receiver treats the value as 250 metres. Which problem is this?

A. A binary addition carry

B. Insufficient colour depth

C. Semantic interoperability failure

D. A guaranteed encryption failure

MC23 — Chapter 6: Strengths & Limitations / Put each computation where it fits

Which is the strongest reason to keep an emergency stop loop local?

A. Cloud servers can never learn models

B. Local models are always perfectly accurate

C. Local computation has unlimited energy

D. Avoid dependence on a variable remote round trip for a strict deadline

MC24 — Chapter 6: Strengths & Limitations / Understand why a policy acted

Which statement best evaluates a model explanation?

A. A colourful heat map proves the model reasons like a human

B. Explanations replace testing on new cases

C. Any explanation proves fairness

D. Its faithfulness, audience and limitations must be assessed

Questions — Open-ended

OE01 — Chapter 1: The Fundamentals / Trace a tiny program

Pip's parcel counter uses the toy instructions 100 LOAD #3; 101 ADD #4; 102 STORE 200; 103 HALT. Start with PC 100, ACC 0 and memory[200] 0. Trace PC, ACC and memory[200] after each instruction; explain immediate values versus addresses; change the ADD operand to #9 and predict the result. Then explain how a source program reaches execution and state two limits of this teaching model. Provide at least 120 words or an equivalently detailed labelled trace plus explanations.

OE02 — Chapter 2: Behaviorism & Behaviorist AI / Update a Q value

Pip tries a route. Old Q = 2, alpha = 0.5, reward = 1, gamma = 0.9 and best next Q = 4. Calculate target, error, change and updated Q. Repeat if the next state is terminal. Explain reward versus policy versus value, why one successful trip is insufficient, and how exploration can be useful without ignoring safety. Provide at least 120 words or equivalently developed calculations and reasoning.

OE03 — Chapter 3: Hardware / Overlap work with a pipeline

A fictional CPU has four one-cycle stages F, D, E, W. Schedule three independent instructions in an ideal pipeline and compare the elapsed cycles with no overlap. Explain throughput versus single-instruction latency. Then discuss what changes if I2 needs I1's result, available only after the end of I1's W, with no forwarding. Finally explain why GPU, RISC or clock-rate labels alone do not establish the best processor for Pip. Provide a labelled schedule and developed explanation.

OE04 — Chapter 4: Networking / Capacity is not responsiveness

A warning travels over a 100 Mb/s connection but can arrive after 500, 50, 5 or 200 ms. Use a constant illustrative speed of 30 m/s to calculate distance travelled during each delay. Explain why this is not total stopping distance, how latency differs from bandwidth and jitter, and why switching from UDP to TCP cannot automatically guarantee the safety deadline. Propose two measurements and one design response. Give at least 120 words or equivalent calculations and analysis.

OE05 — Chapter 5: Applications / Networking for a behaviour-based agent

A receiver expects robot-7, units m, policy_version 3, authenticated input, maximum age 200 ms and sequence greater than last accepted sequence 41. Process A: sequence 42, age 50 ms; B: sequence 43, age 600 ms; C: sequence 42, age 60 ms. All other fields match. Give each decision and receiver state. Explain identity, freshness, replay and unit checks, why stale rejected input must not advance accepted state, and how delayed rewards should be associated with actions. Include at least 120 words or equivalent explicit decision reasoning.

OE06 — Chapter 6: Strengths & Limitations / Put each computation where it fits

Choose a design for an 80 ms local control task with 512 MB RAM and a 6 W budget. Small local model: 30 ms, 180 MB, 4 W, 92% task accuracy. Large local: 90 ms, 700 MB, 9 W, 96%. Remote: 20 ms compute plus 100–300 ms network round trip, 98%, with image uploads. These are fictional supplied measurements. Evaluate each option, propose a split between local control and fleet analytics, and discuss thermal testing, privacy, fairness and responsibility. Give at least 150 words or an equivalent decision matrix with developed reasoning.

Answer key — read after attempting all questions

True/False answers and corrections

TF01: True

The most significant bit has weight −128; the remaining seven bits can contribute 0 through 127.

TF02: False

Unicode assigns code points; encodings such as UTF-8 use variable numbers of bytes. A displayed grapheme can also contain more than one code point.

TF03: False

It specifies clock cycles per second. Instruction mix, stalls, issue width and microarchitecture affect completed work.

TF04: True

The operand is a destination address; the stored value is the accumulator's current content.

TF05: False

It is reactive control. RL requires an experience-based update to value estimates or a policy.

TF06: True

The objective normally concerns cumulative reward, not only the next reward.

TF07: True

The episode ends, so no later return is bootstrapped from that terminal state.

TF08: False

Inference can use fixed parameters. Online learning is a separate design choice.

TF09: False

Both are normally volatile. SRAM avoids periodic refresh while powered; this does not make it non-volatile.

TF10: True

The finish cycles are 4, 5 and 6 under the stated equal-stage, no-stall assumptions.

TF11: False

Parallelism only helps workloads that expose suitable independent work; transfer overhead and dependencies also matter.

TF12: True

This hysteresis prevents small fluctuations around a single threshold from repeatedly toggling the actuator.

TF13: False

Propagation, queues, endpoint processing and other factors can create delay even on a high-capacity link.

TF14: True

Legacy shared half-duplex Ethernet used collision detection. Congestion can still occur on modern switched networks.

TF15: False

A signature supports origin and integrity relative to a trusted verification key. Encryption is a separate confidentiality mechanism.

TF16: True

Name resolution is one stage; network reachability, server availability or application errors can still prevent a response.

TF17: False

That 120 ms response misses the stated deadline. An average hides individual delays.

TF18: True

Authentication and freshness answer different questions. Replay, sequence and timing checks may still be needed.

TF19: False

The reward must be associated with its actual transition or episode; arrival order alone can associate it incorrectly.

TF20: True

The actuator can stall, saturate or fail; outcome feedback is needed to check the physical result.

TF21: False

Storage is only one constraint. Weights, activations, runtime buffers, sustained power and deadline performance also matter.

TF22: True

Correct use of exchanged information requires shared meaning, not merely delivery of bytes.

TF23: False

Cloud-dependent tasks can fail when connectivity is unavailable. Local fallbacks and placement must match task requirements.

TF24: True

Explanation methods have assumptions and limits. Evaluate what the explanation establishes for its audience.

Multiple-choice answers and reasoning

MC01: D

Ki means 1,024: 2 × 1,024 = 2,048 bytes. Multiplying by 8 would convert bytes to bits.

MC02: D

16 colours need 4 bits per pixel. 500 × 200 × 4 / 8 = 50,000 bytes.

MC03: C

MAR contains the address. MDR carries the transferred value or instruction; CIR holds the current instruction.

MC04: A

LOAD sets ACC to 3 and ADD makes 12. STORE copies 12 to address 200.

MC05: B

A policy governs action choices. Reward scores outcomes; the environment supplies transitions and feedback.

MC06: C

Target = 1 + 0.9 × 4 = 4.6; error = 2.6; change = 1.3; new Q = 2 + 1.3 = 3.3.

MC07: A

Always exploiting early estimates can prevent learning about alternatives. Exploration addresses uncertainty, subject to costs and safety constraints.

MC08: C

0.8 × 10 + 0.2 × (−5) = 8 − 1 = 7. This is an expectation, not the outcome of every trial.

MC09: A

DRAM stores leaking charge; SRAM uses bistable circuits. Both normally lose data without power.

MC10: A

For k equal stages and n instructions the ideal time is k + n − 1 = 4 + 5 − 1 = 8 cycles.

MC11: C

NOT 1 = 0, then 1 AND 0 = 0. The obstacle prevents movement even with a healthy battery.

MC12: D

Independent pixel operations can use the same instruction over multiple data elements. This does not imply every image workload has no dependencies.

MC13: B

50 ms = 0.05 s; 30 × 0.05 = 1.5 m. This excludes braking and other delays.

MC14: D

The Internet layer handles IP addressing and routing. TCP is transport; HTTP is application.

MC15: B

Validation tests conformance to rules. A plausible but factually wrong value can still pass.

MC16: B

Signature algorithms use a private signing key and a matching public verification key. Trust in the public key's identity is also necessary.

MC17: B

60 ms is below 80 ms. A good average does not make a particular 90 ms response timely.

MC18: B

Age exceeds the threshold. A rejected packet must not advance the accepted-sequence state.

MC19: A

The valve actuator creates physical change. The sensor measures a condition; software selects a command.

MC20: A

Learning needs the correct state/action/outcome association. Network arrival order may differ from event order.

MC21: C

180 MB ≤ 512 MB and 4 W ≤ 6 W. Timing and accuracy still need separate evaluation.

MC22: C

The systems disagree about meaning/units despite exchanging a value. A schema and unit agreement are needed.

MC23: D

Local control can remove a network dependency from a timing-critical path, while still requiring reliable hardware and safety design.

MC24: D

An explanation should be judged by what it validly reveals to a user, not by its visual polish. It complements, rather than replaces, evaluation.

Open-ended expected points and sample responses

Valid alternatives receive credit when justified with correct assumptions and evidence. These sample responses demonstrate one acceptable line of reasoning.

OE01 — expected points (10 marks)

• Correct original trace: ACC 3, 7, 7, 7; memory[200] becomes 7 at STORE; final PC 104 under the stated fetch rule (3 marks).

• Modified result 12, with reasoning from 3 + 9 (1 mark).

• Distinguishes # immediate values from a memory address (2 marks).

• Explains assembler/compiler/interpreter or hybrid implementation without a language-wide false dichotomy (2 marks).

• States limits such as fixed instruction width, invented ISA/registers and simplified timing (2 marks).

Sample response: The program begins with PC 100. Fetching LOAD advances PC to 101 and execution places literal 3 in ACC. Fetching ADD advances PC to 102, then ACC becomes 7. STORE advances PC to 103 and copies the value 7 to address 200. HALT is fetched, advancing PC to 104 in this teaching model, and stops execution. The # mark denotes a literal value; 200 in STORE is a location rather than the result. Replacing ADD #4 with ADD #9 makes ACC and the stored result 12. In a real workflow, a compiler may produce machine code or an intermediate form, an assembler resolves assembly operations/labels, and a runtime may interpret or compile bytecode. These stages depend on the implementation. The example is an invented one-word ISA with simplified timing; real processors have architecture-specific instructions, state and control-flow rules.

OE02 — expected points (10 marks)

• Non-terminal calculation: target 4.6, error 2.6, change 1.3, updated Q 3.3 (3 marks).

• Terminal calculation sets future term to zero, giving target 1 and new Q 1.5 (2 marks).

• Accurately distinguishes reward, policy and value (2 marks).

• Explains uncertainty and evidence/exploration rather than guaranteed success (2 marks).

• Includes a reasonable safety constraint or simulated-training limitation (1 mark).

Sample response: The non-terminal target is $1 + 0.9 \times 4 = 4.6$. Subtracting old Q = 2 gives temporal-difference error 2.6. The learning rate multiplies that by 0.5, so the change is 1.3 and the updated value is 3.3. If the next state is terminal, no future value is bootstrapped: target = 1, error = −1, change = −0.5 and updated Q = 1.5. Reward is the numerical feedback for an interaction; policy determines action choices; Q estimates expected return from a state–action choice. One trip is a noisy sample, not proof of a safe or optimal route. Exploration can reveal an untested route with better return, but its cost and risk matter. A robot can use restricted safe actions and simulation before physical trials, while still testing whether a policy trained in simulation transfers to the real operating domain.

OE03 — expected points (10 marks)

• Ideal schedule completes I1/I2/I3 on cycles 4/5/6; sequential total 12 (3 marks).

• Distinguishes four-stage instruction latency from overall throughput (2 marks).

• Places I2 E no earlier than cycle 5 under the explicit no-forwarding assumption, with consistent downstream waiting (2 marks).

• Identifies workload, dependencies, memory and power/timing measures (2 marks).

• Avoids universal performance claims from RISC/CISC/GPU/GHz labels (1 mark).

Sample response: In the ideal schedule I1 occupies F1, D2, E3, W4; I2 occupies F2, D3, E4, W5; I3 occupies F3, D4, E5, W6. The batch finishes in six cycles rather than twelve sequential stage cycles. Each instruction still passes through four stages, so pipelining improves completion rate after filling rather than reducing an instruction to one stage. Under the separate no-forwarding assumption, I2 cannot execute before cycle 5 because I1's result becomes available only after cycle 4. It writes in cycle 6. With instructions kept in order, I3 waits behind I2, executes in cycle 6 and writes in cycle 7. Actual CPU forwarding and hazard rules may differ. Pip's processor choice should be evaluated on its real workload, sustained latency, memory traffic, parallelism and energy. A GPU helps suitable parallel arithmetic; an instruction-set label or clock frequency cannot by itself predict application performance.

OE04 — expected points (10 marks)

• Converts ms to s and obtains 15, 1.5, 0.15 and 6 m respectively (3 marks).

• Separates delay distance from actuator/braking and other response terms (2 marks).

• Accurately defines bandwidth, latency and jitter (2 marks).

• Explains that reliable ordered transport does not promise a fixed deadline (1 mark).

• Proposes relevant worst-tail/end-to-end measurements and a local/fallback response (2 marks).

Sample response: The delays are 0.5, 0.05, 0.005 and 0.2 seconds. Multiplying by 30 m/s gives delay distances of 15, 1.5, 0.15 and 6 metres. These figures measure only travel during the stated communication delay. They omit processing after receipt, actuator response and braking, so they are not total stopping distances. Bandwidth is a link's capacity; latency is delay; jitter is variation in delay. A high-capacity link can still suffer queues or long paths. TCP can provide ordered reliable delivery while retransmission and head-of-line waiting add time; it does not automatically guarantee the application's deadline. Measure end-to-end command age under load, including high-percentile or worst observed delay, and measure complete sensor-to-actuator response under relevant failures. Keep timing-critical stopping local when appropriate and define safe behaviour on late, missing or invalid remote commands.

OE05 — expected points (10 marks)

• A accepts and updates to 42; B rejects stale and remains 42; C rejects duplicate and remains 42 (3 marks).

• Distinguishes authentication, identity/units/version, freshness and replay checks (3 marks).

• Explains accepted-state consistency and safe local response on rejection (2 marks).

• Associates reward to its original transition/episode and notes toy authentication limitation (2 marks).

Sample response: Packet A satisfies the identity, units, version, age and sequence requirements, so the receiver accepts it and stores last accepted sequence 42. Packet B has a newer sequence but is 600 ms old, beyond the 200 ms limit; it is rejected as stale and the accepted sequence remains 42. Packet C is recent enough but repeats sequence 42, so it is rejected as a duplicate without changing state. Authentication concerns who may send a protected message; expected device identity and policy version route it to the correct interpretation. Units ensure the numerical value has the intended meaning. Freshness and sequence/session rules address delayed or replayed messages. A rejected packet must not corrupt the bookkeeping that describes accepted commands. The local controller should use its defined safe behaviour. Delayed rewards belong to identified transitions or episodes, not merely the latest action. The worksheet's authentication Boolean assumes an upstream check and is not a cryptographic protocol.

OE06 — expected points (10 marks)

• Correctly checks all three hard budgets and rejects local large (3 marks).

• Computes remote 120–320 ms and rejects it for the 80 ms task (2 marks).

• Justifies local control and slower remote analytics with a connectivity fallback (2 marks).

• Addresses sustained thermal/timing validation and relevant accuracy/impact testing (1 mark).

• Includes concrete privacy and accountability measures, recognising supplied accuracies are not proven deployment safety (2 marks).

Sample response: The small local model fits all stated limits: 30 ms is below 80 ms, 180 MB is below 512 MB and 4 W is below 6 W. The large local model misses the deadline and exceeds both memory and power budgets despite its better supplied accuracy. The remote option needs 20 + 100 to 20 + 300 ms, or 120–320 ms, and therefore fails the local deadline throughout the stated range. Its 98% figure cannot compensate for this hard constraint. I would keep the immediate control loop and a defined safe stop locally, and use remote processing for slower fleet analysis or training when connectivity is available. I would test warm-device sustained timing, memory peaks, adverse sensor conditions and differences across the intended environments. Image uploads need a legitimate purpose, minimised content, limited retention and access controls. A named team or operator must own monitoring, updates, incident response and appeals or correction where people are affected. These fictional average accuracy figures are useful comparison inputs, not evidence of a safe deployment.

Review and provenance

This question set was AI-assisted and checked for coverage, arithmetic, answer alignment and document completeness on 7 November 2026. It has not been recorded as independently attempted by another team member. That mandatory human review remains pending and must be documented honestly. Questions are original teaching material based on the module; chapter references and resource attribution are included in the complete Study Guide and Markdown snapshot.